



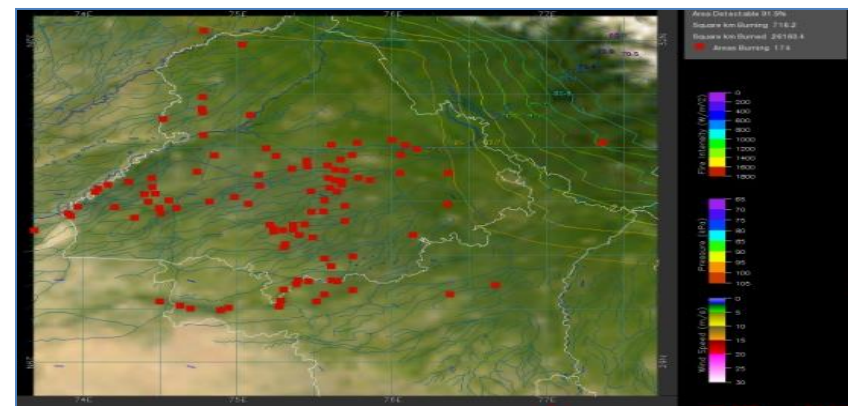
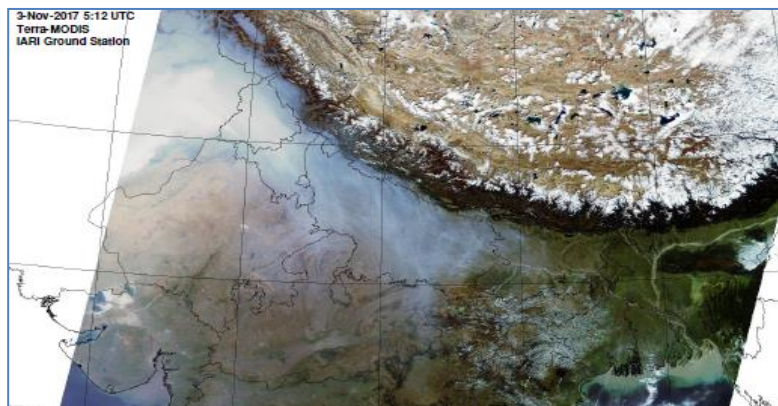
Bulletin No. 28

Events Date:
12-Oct-2023

Issued on:
12-Oct-2023



MONITORING PADDY RESIDUE BURNING IN INDIA USING SATELLITE REMOTE SENSING DURING 2023



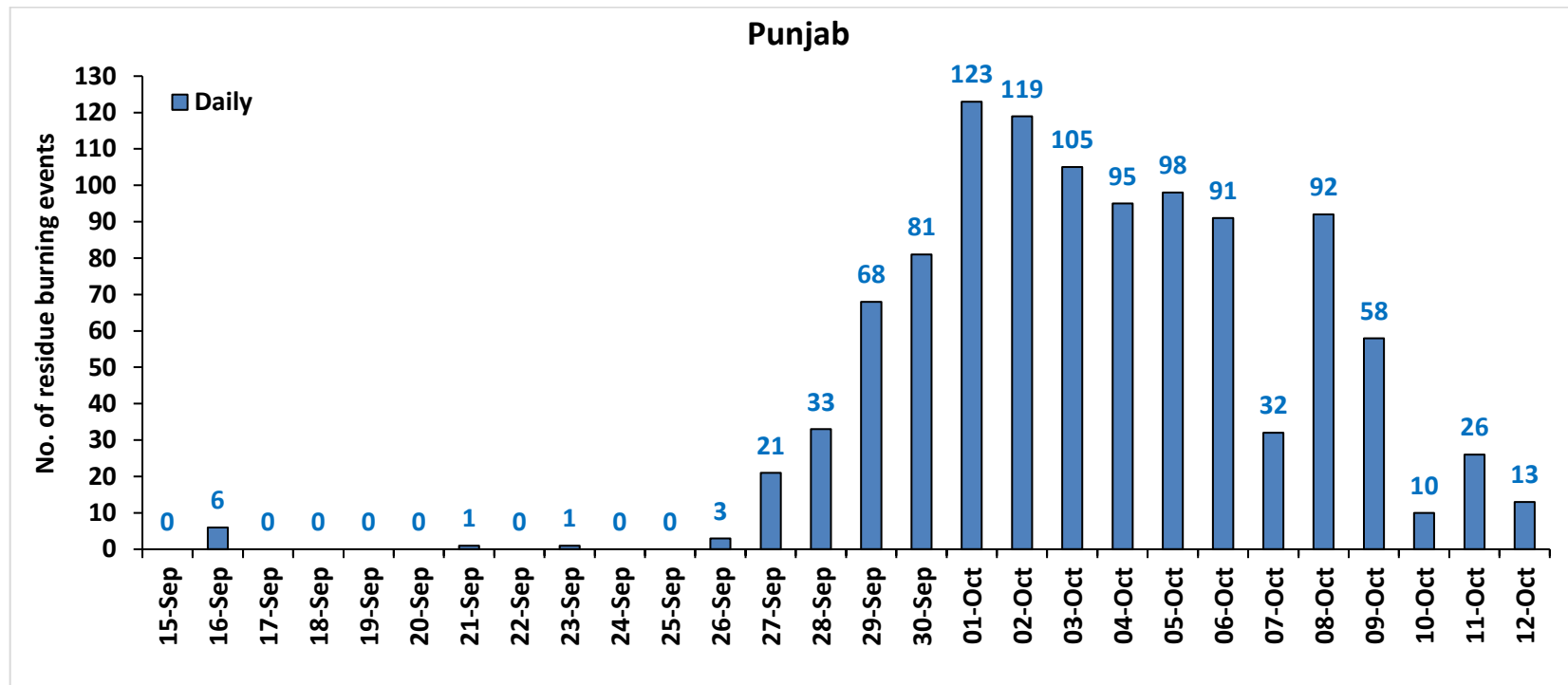
Consortium for Research on Agroecosystem Monitoring and Modeling from Space (CREAMS) Laboratory,
Division of Agricultural Physics, ICAR – Indian Agricultural Research Institute, New Delhi – 110012

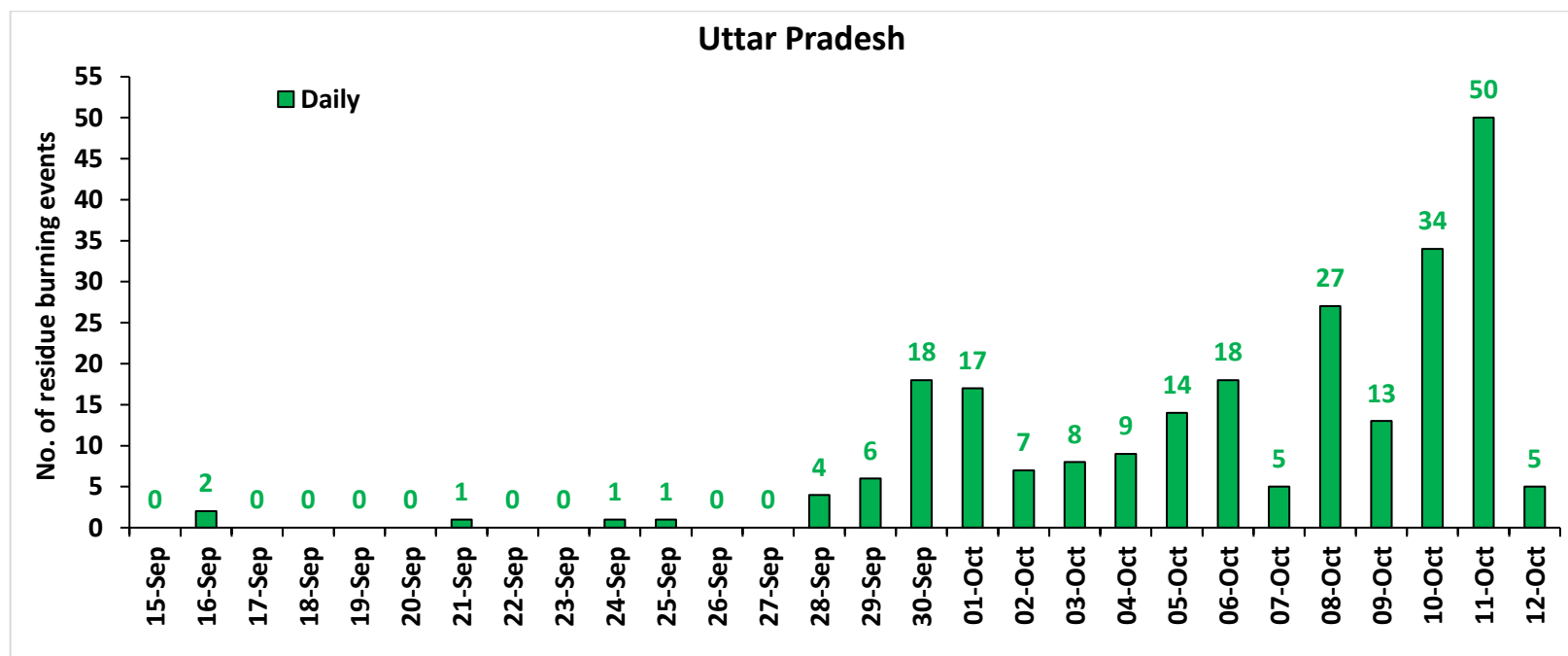
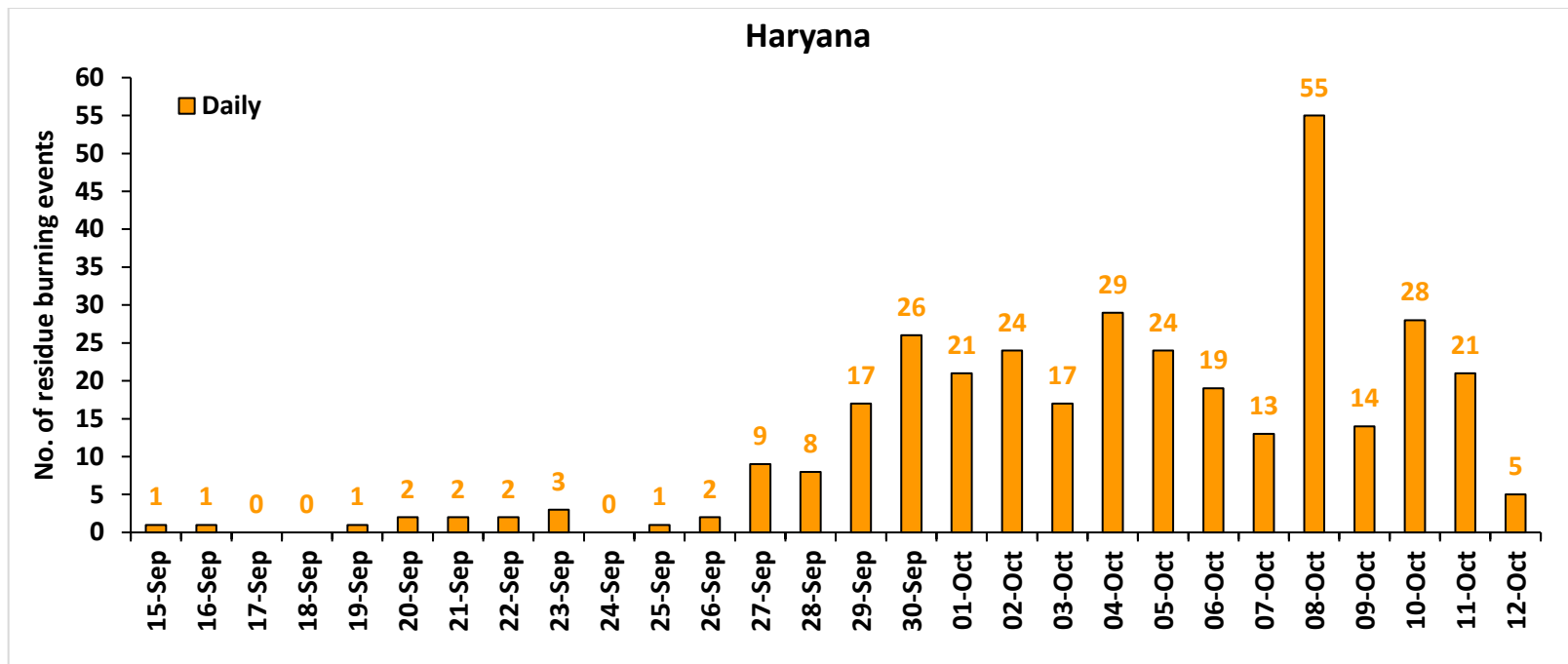
<http://creams.iari.res.in>

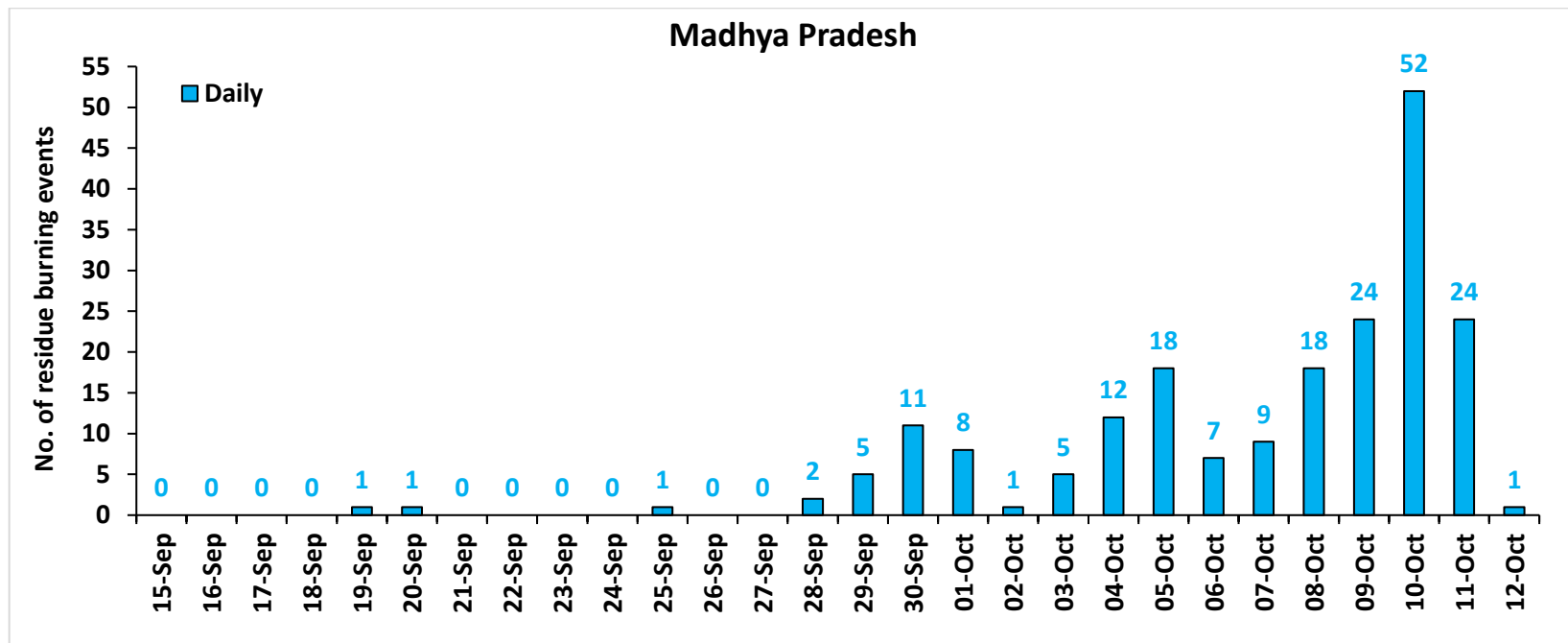
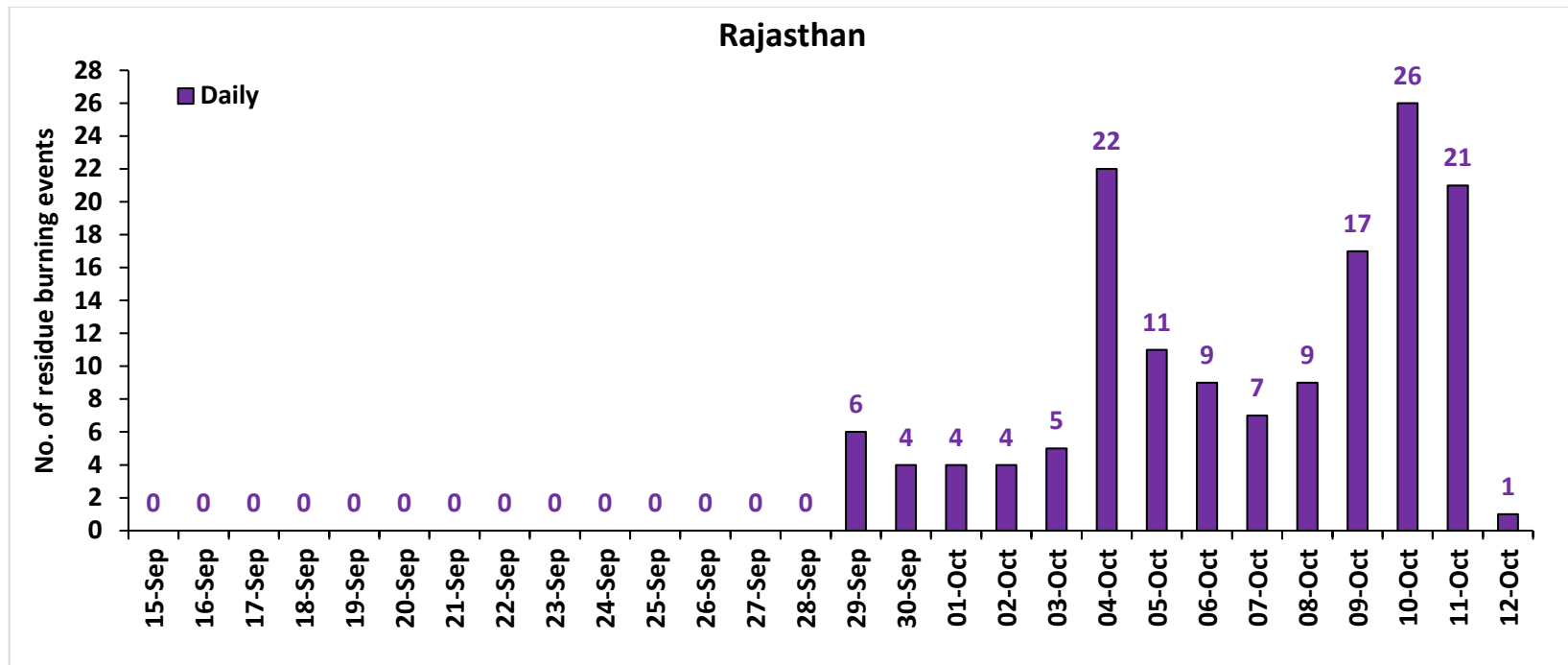
Highlights for 12-Oct-2023

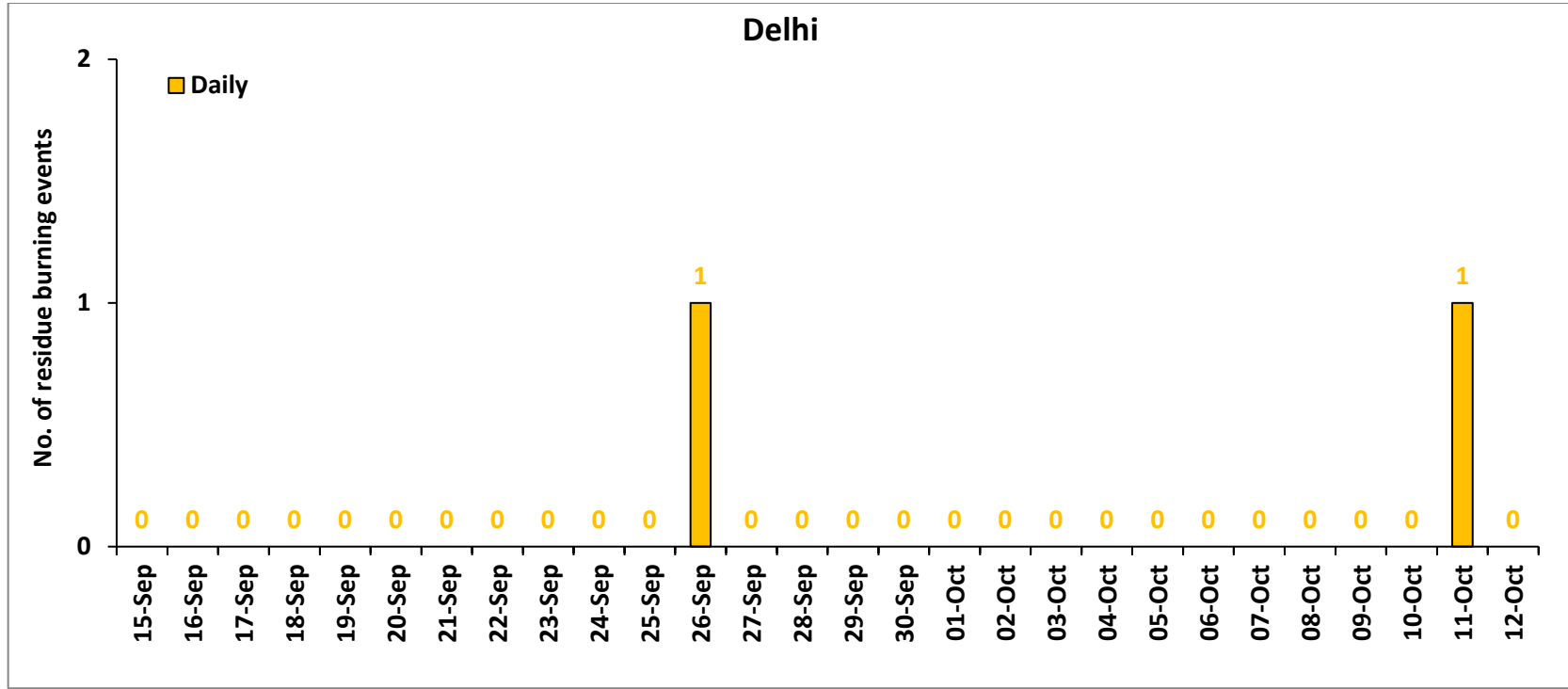
- The active fire events due to rice residue burning were monitored using satellite remote sensing, following the "Standard Protocol for Estimation of Crop Residue Burning Fire Events using Satellite Data".
- Satellites detected **25** residue burning events in the six study States on 12-Oct-2023.
- The state wise events detected on 12-Oct-2023 were **13** in Punjab, **05** in Haryana, **05** in UP, **00** in Delhi, **01** in Rajasthan and **01** in MP.
- Total **2009** burning events were detected in the six states between 15-Sept-2023 and 12-Oct-2023, which are distributed as **1076**, **345**, **240**, **02**, **146** and **200** in Punjab, Haryana, UP, Delhi, Rajasthan and MP, respectively.

Temporal distribution of rice residue burning events for the study States in 2023

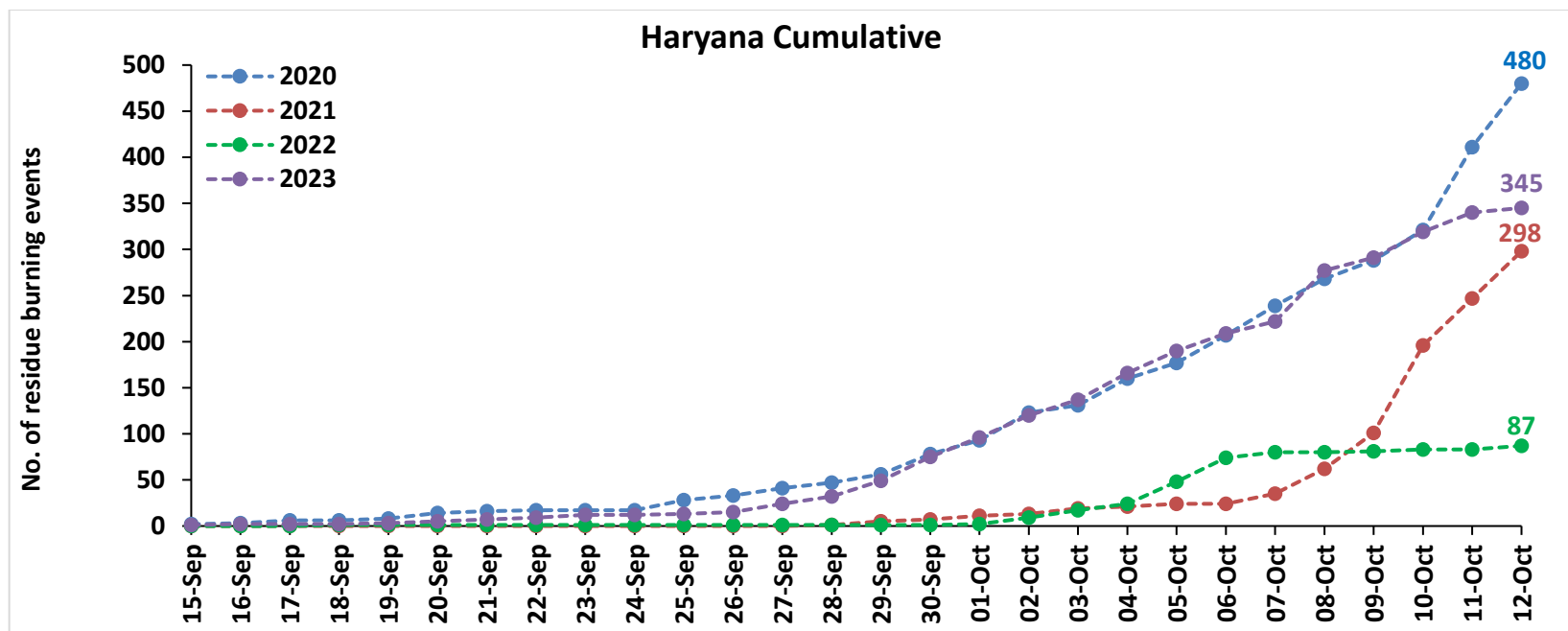
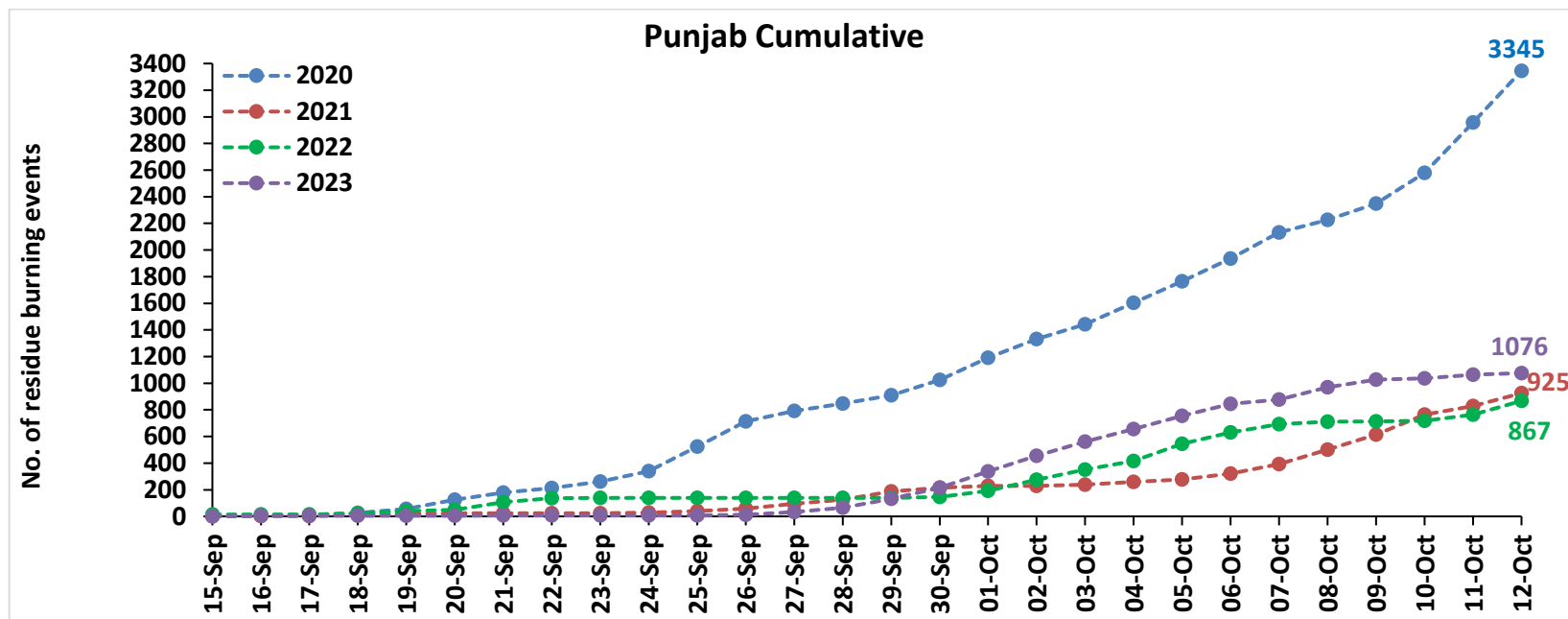


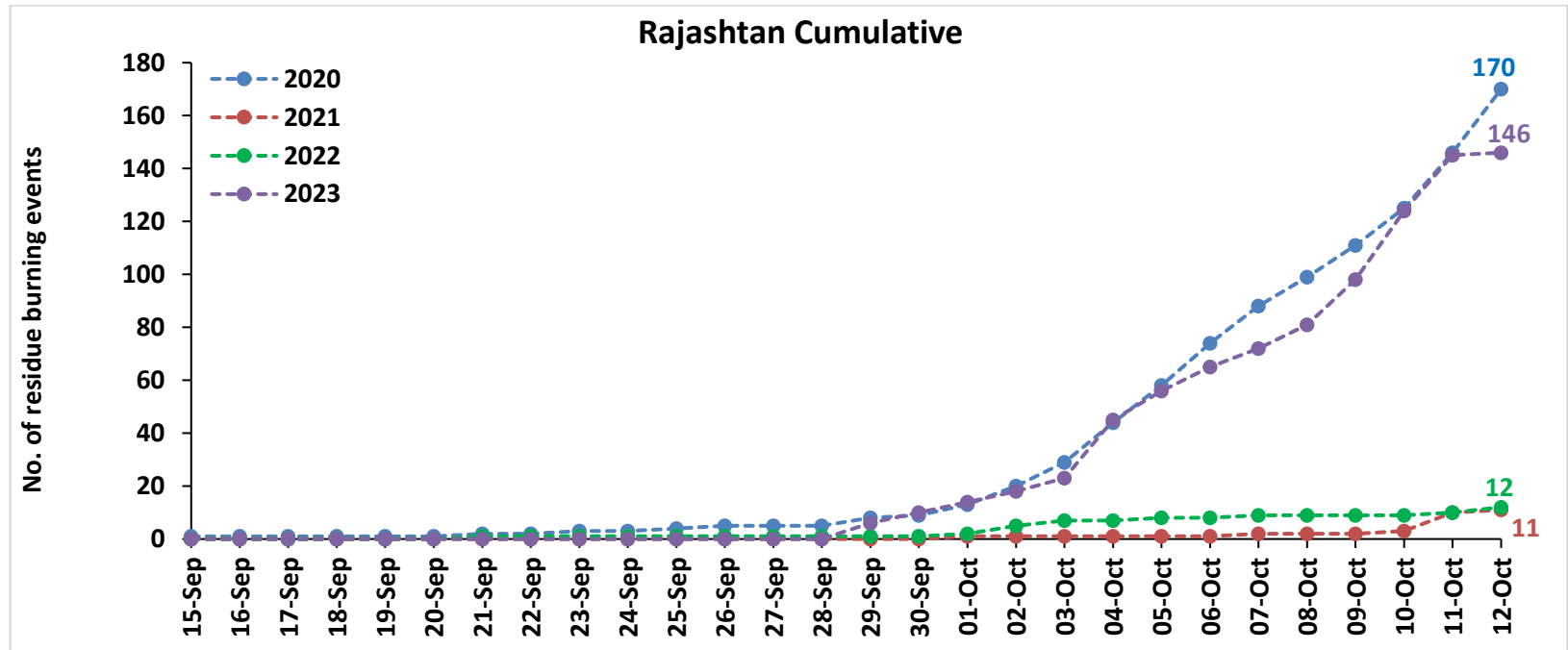
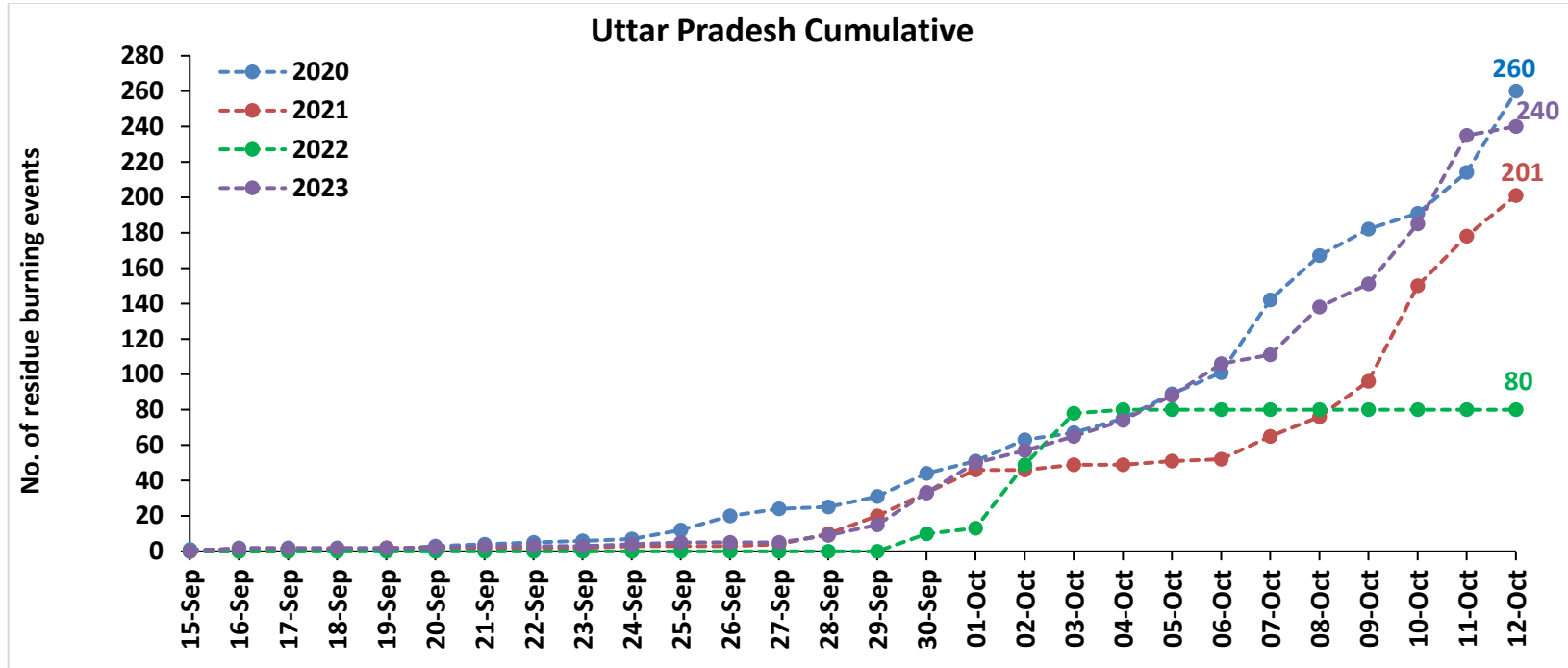


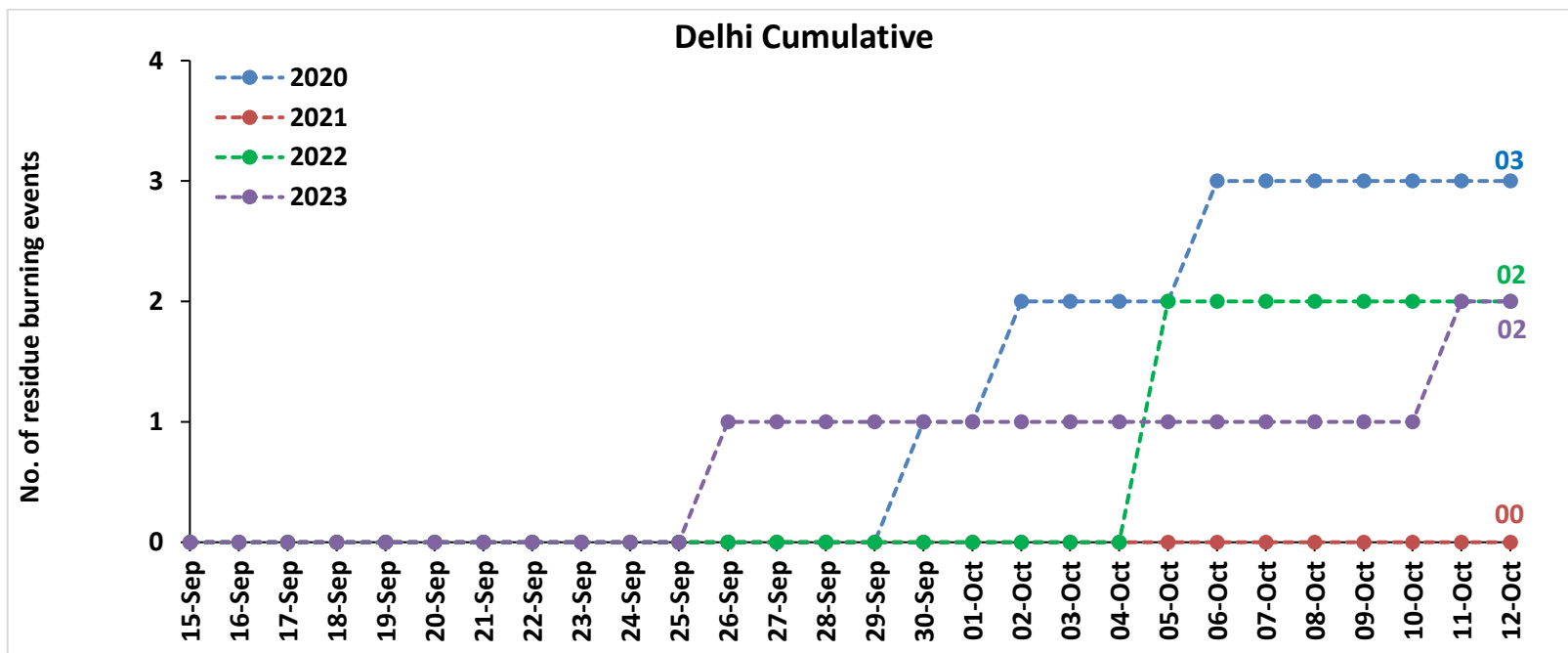
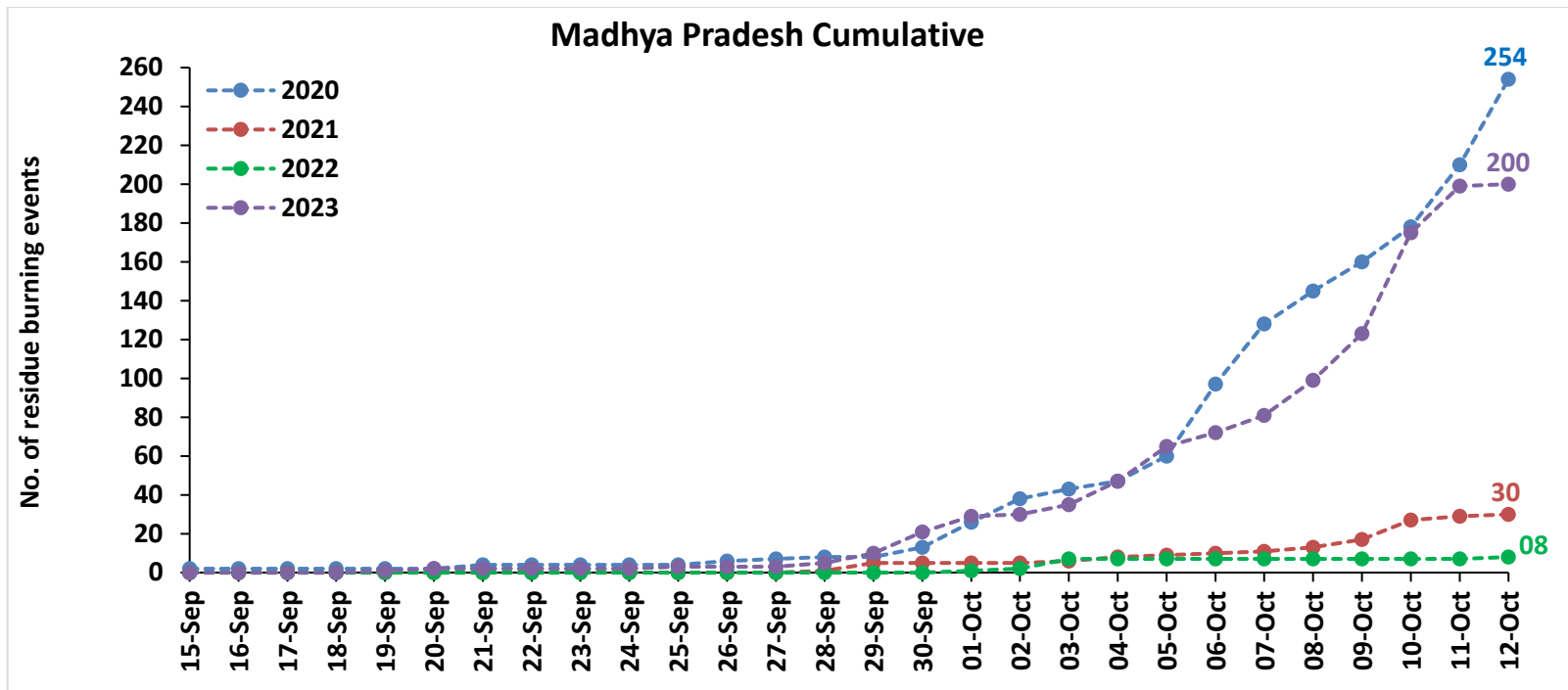




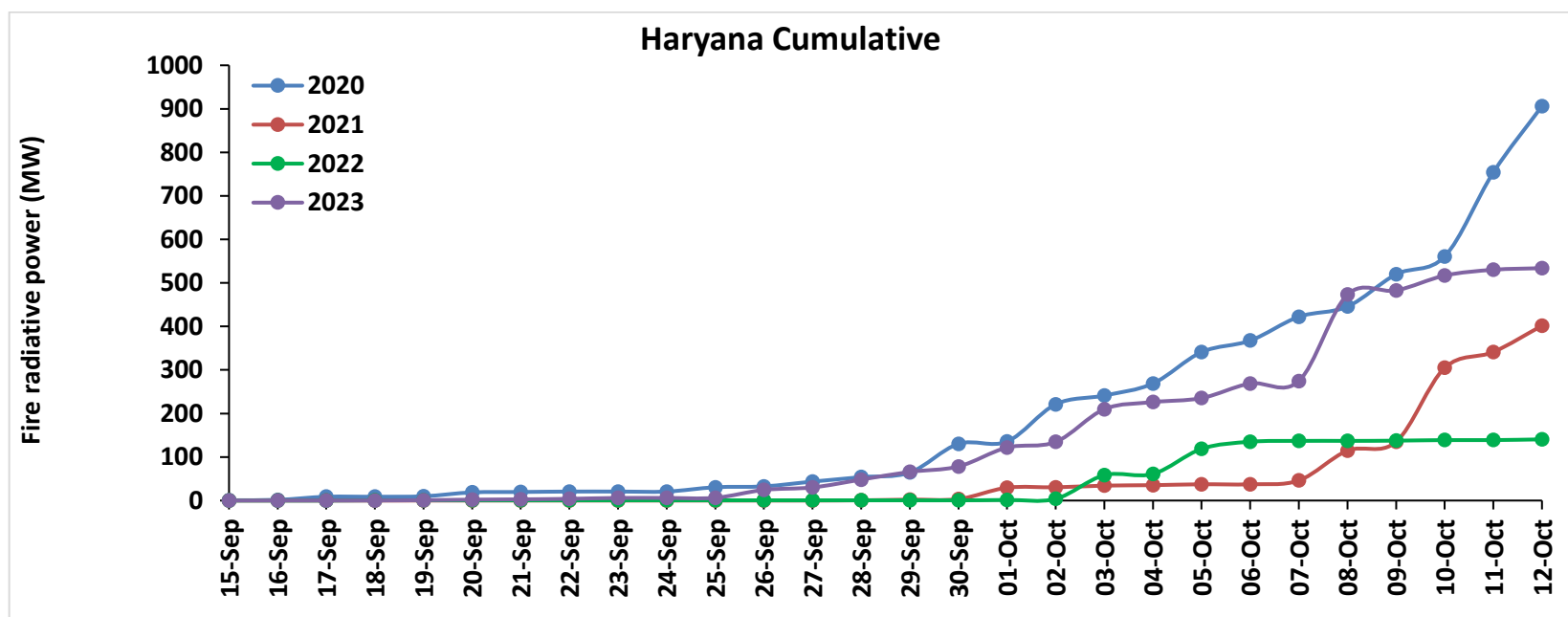
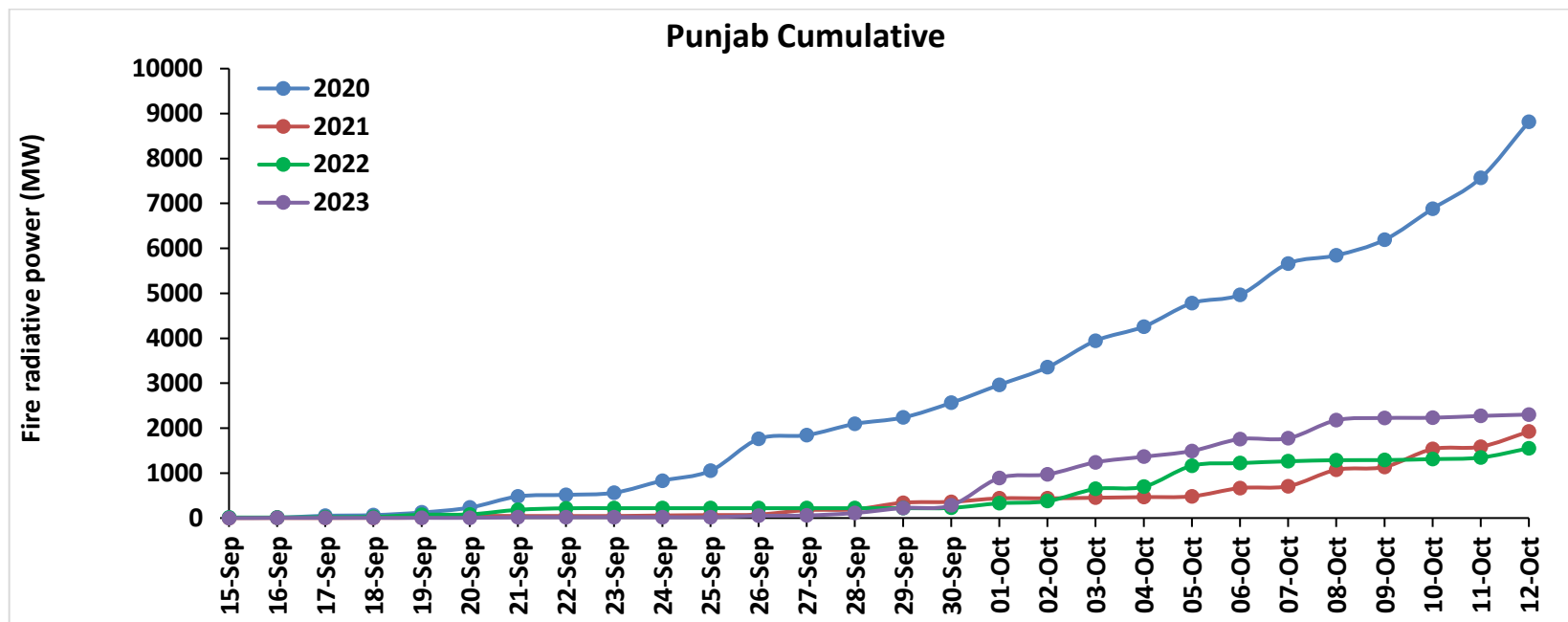
Comparison of residue burning events in current year (2023) with previous years (2020, 2021 and 2022) for the study State

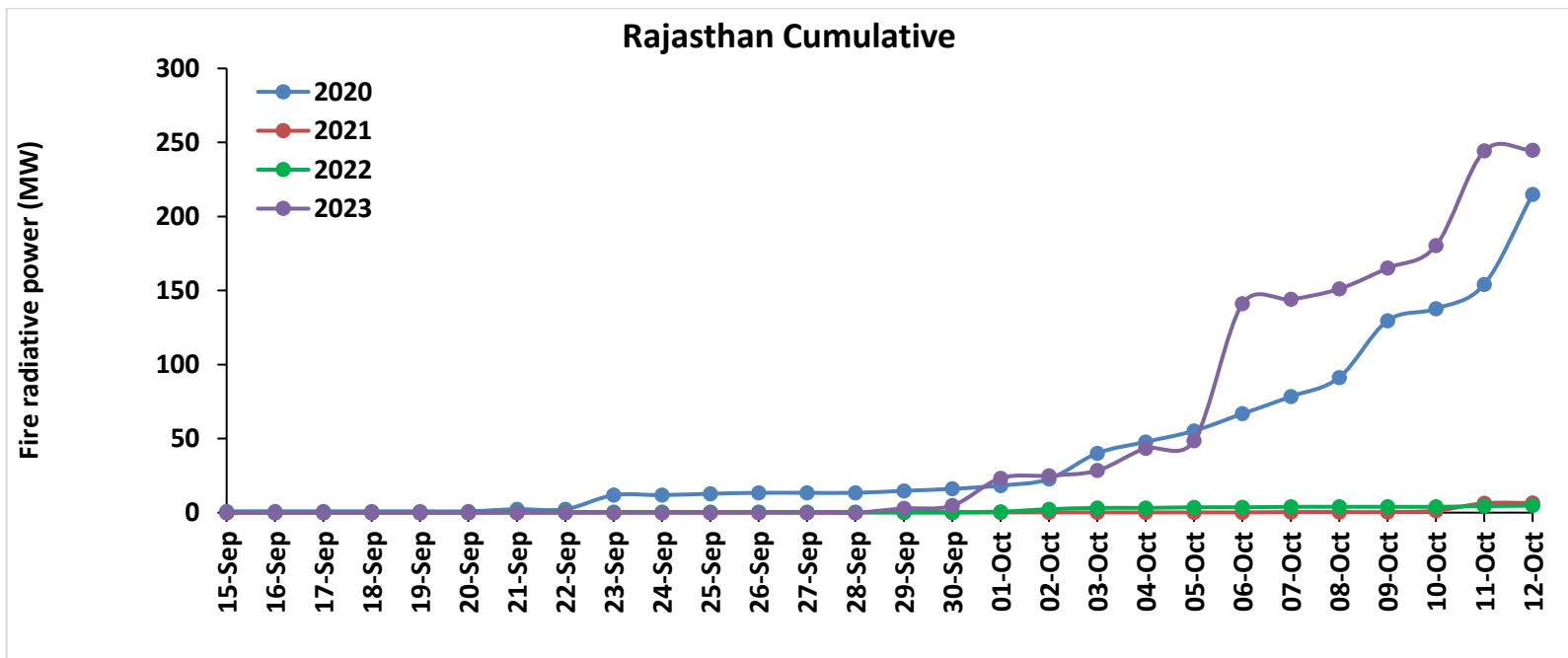
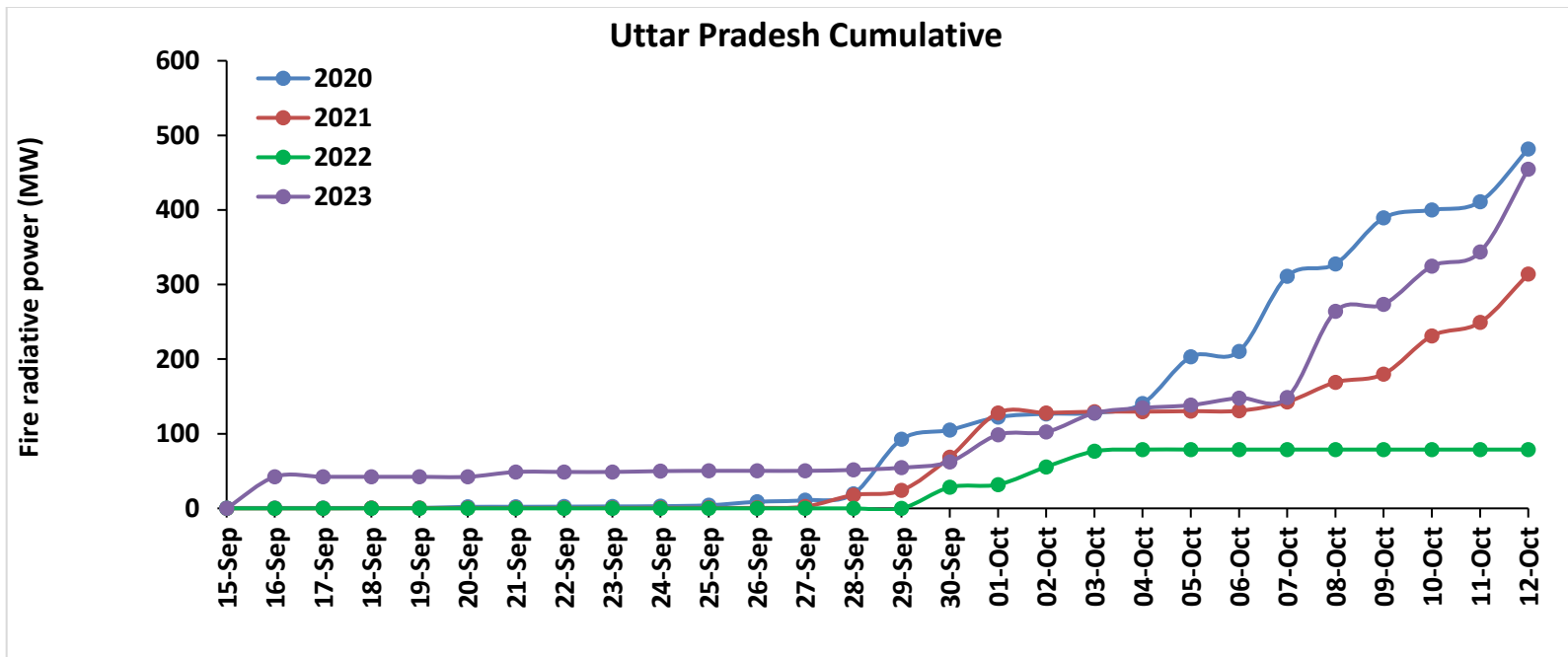


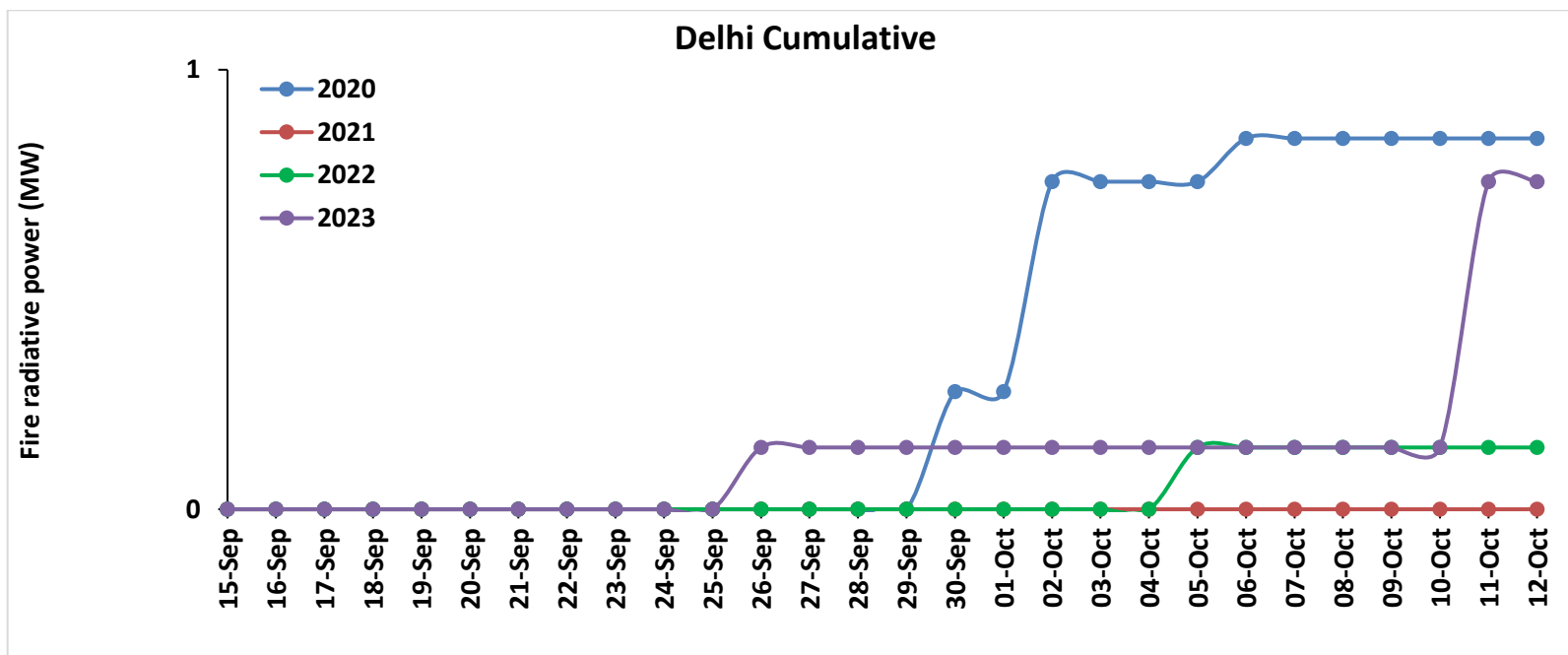
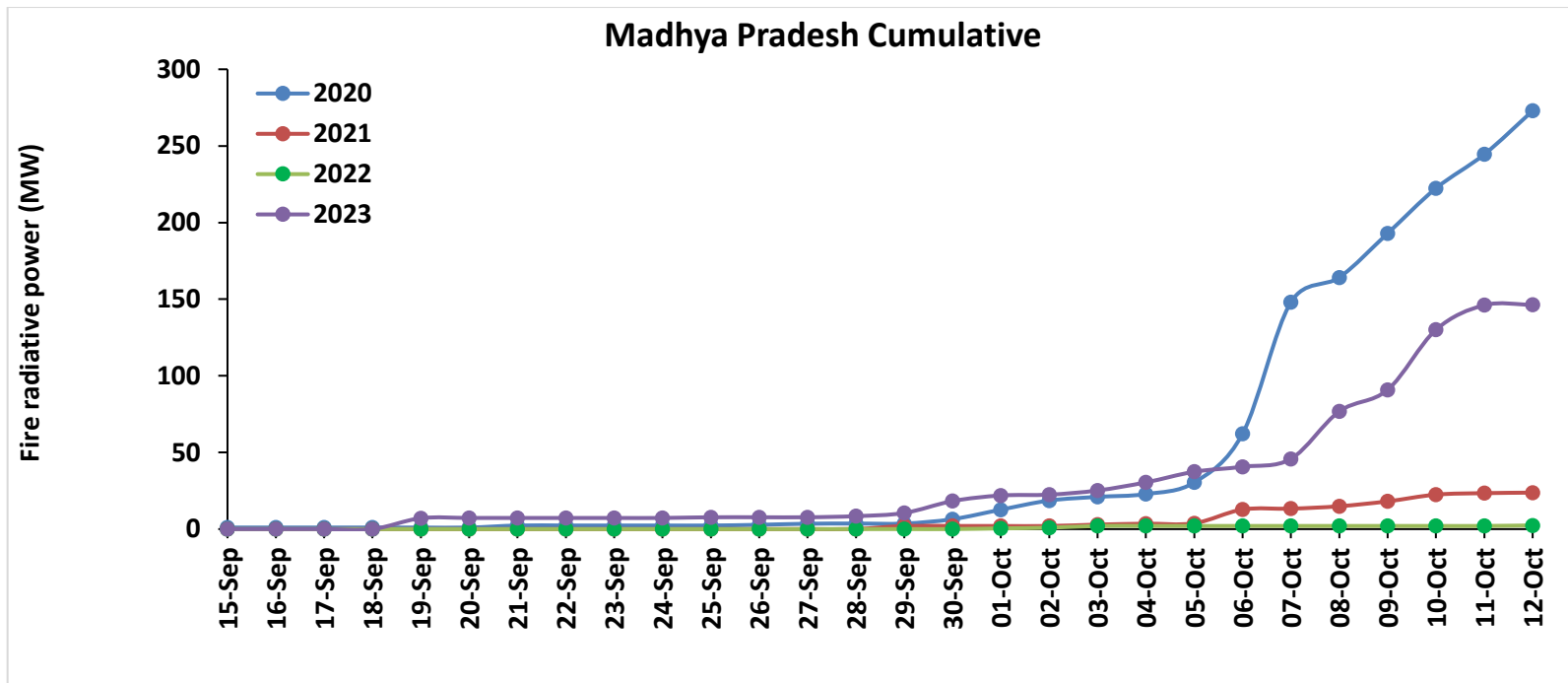




Comparison of residue burning fire power in current year (2023) with previous years (2020, 2021 and 2022) for the study States

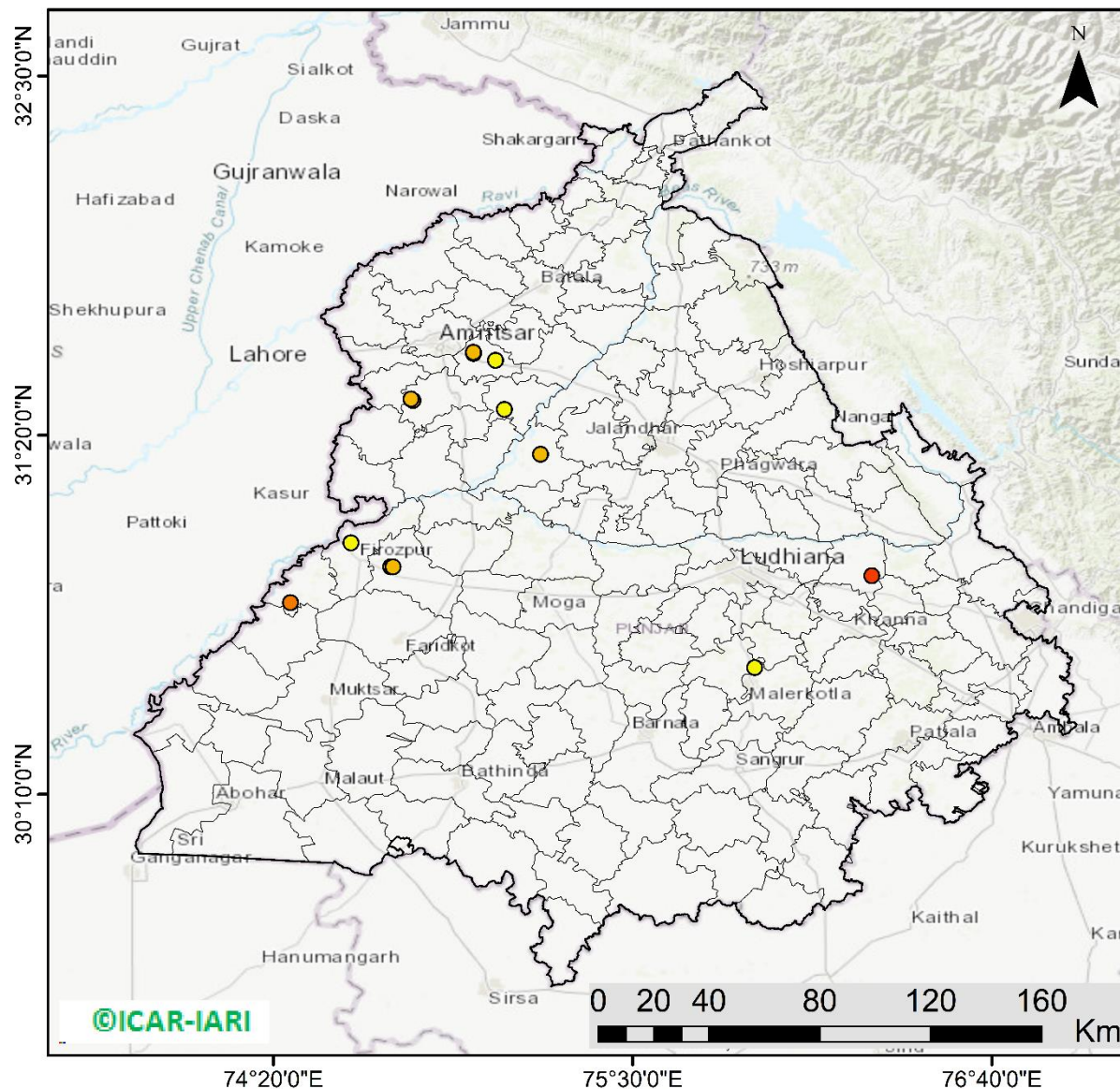






Punjab

RICE RESIDUE BURNING IN PUNJAB



13 burning events
detected in Punjab during
(12th Oct 2023)

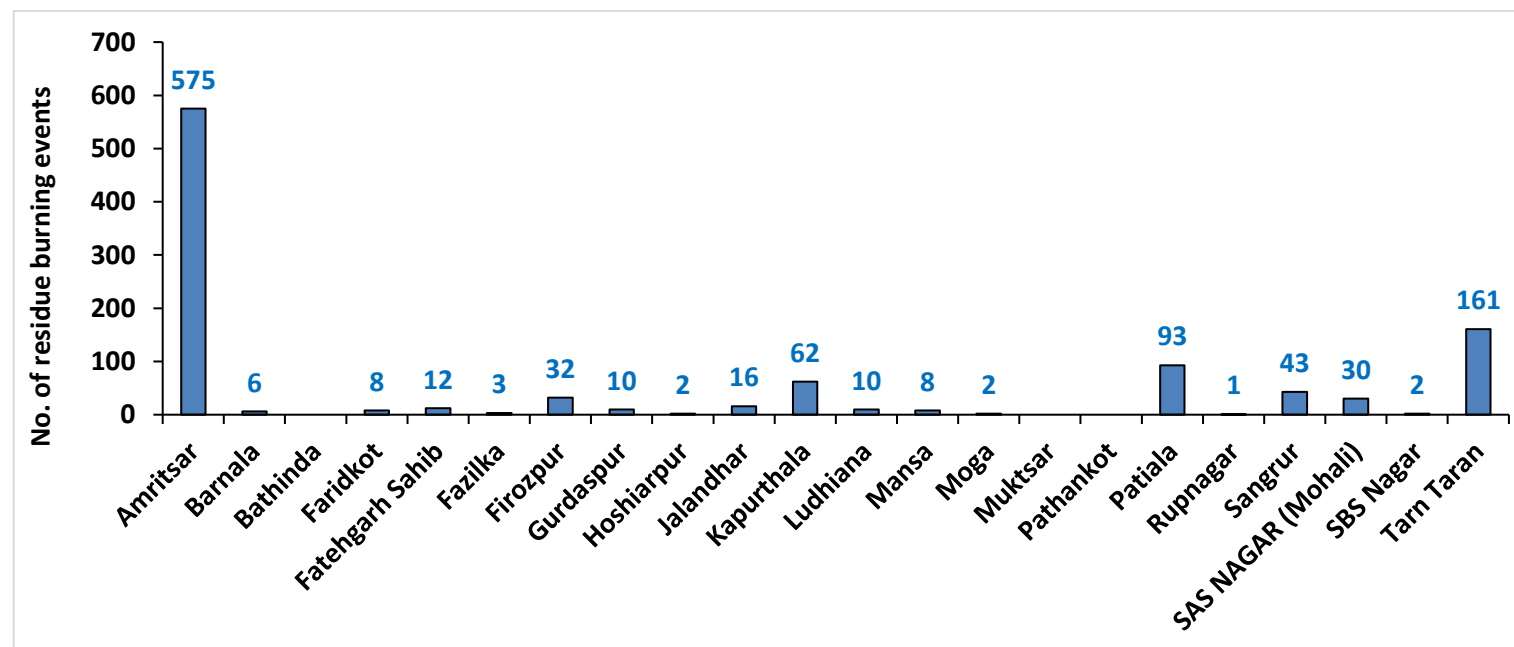
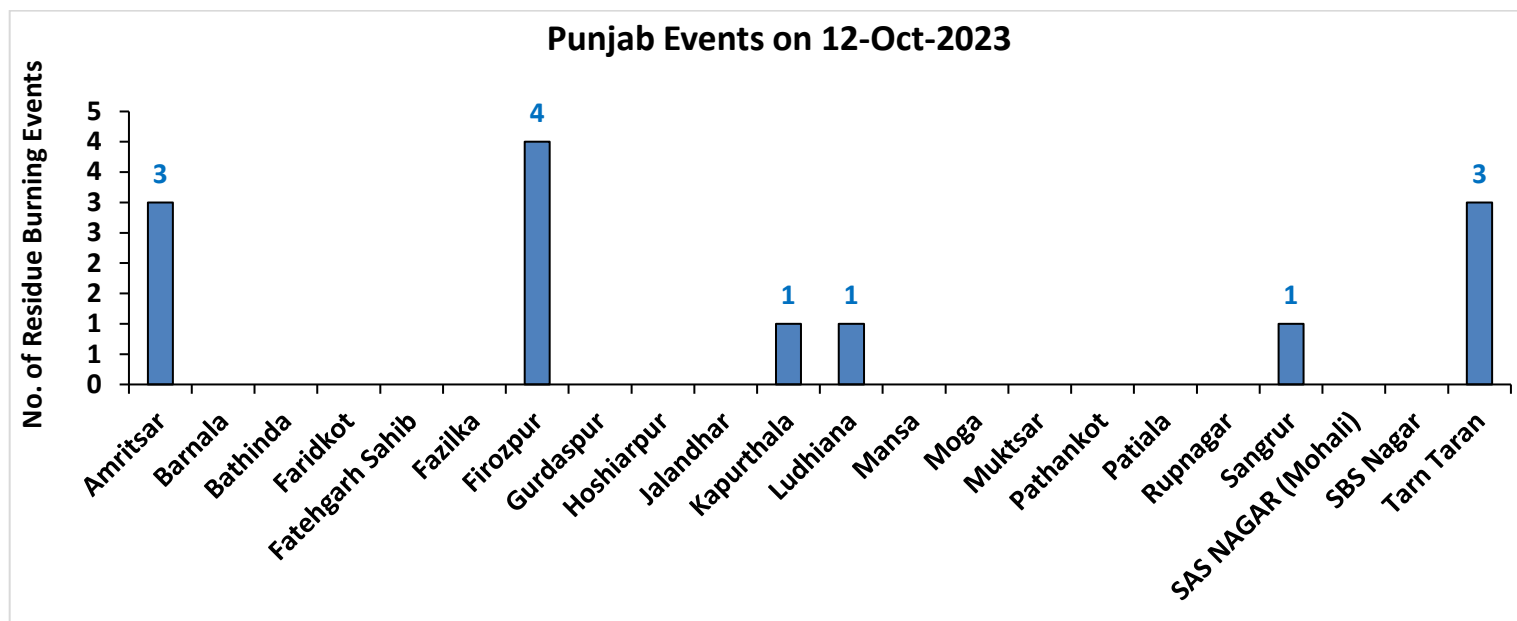
Fire Intensity (W/m²)

- 0 - 5
- 6 - 10
- 11 - 15
- 16 - 20
- >20



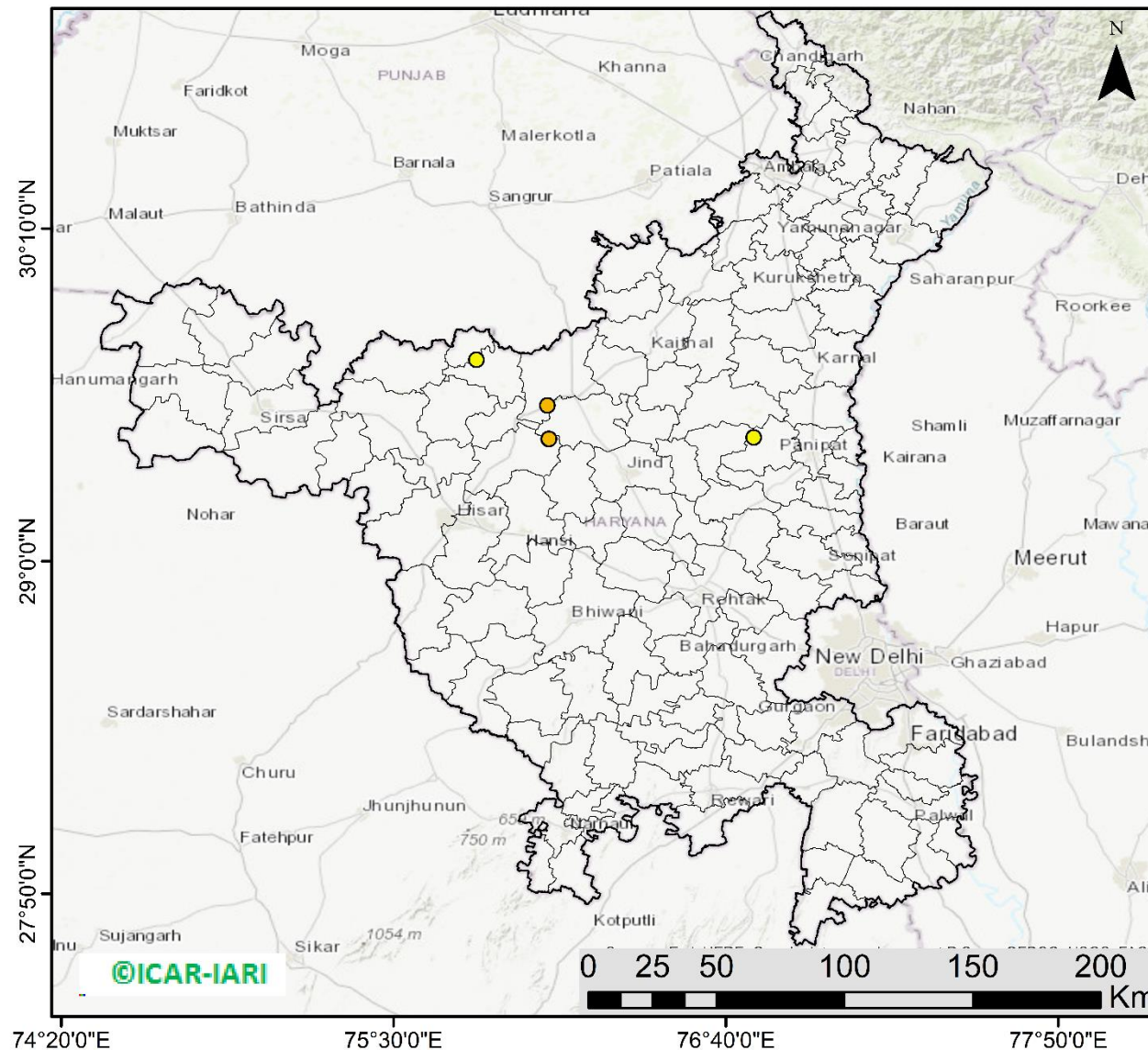
<http://geoportal.icar.gov.in:8080/geoexplorer/composer/>

District-wise cumulative number of residues burning events in Punjab (15Sep- 12Oct-2023)



Haryana

RICE RESIDUE BURNING IN HARYANA



05 burning events
detected in Haryana on
(12th Oct 2023)

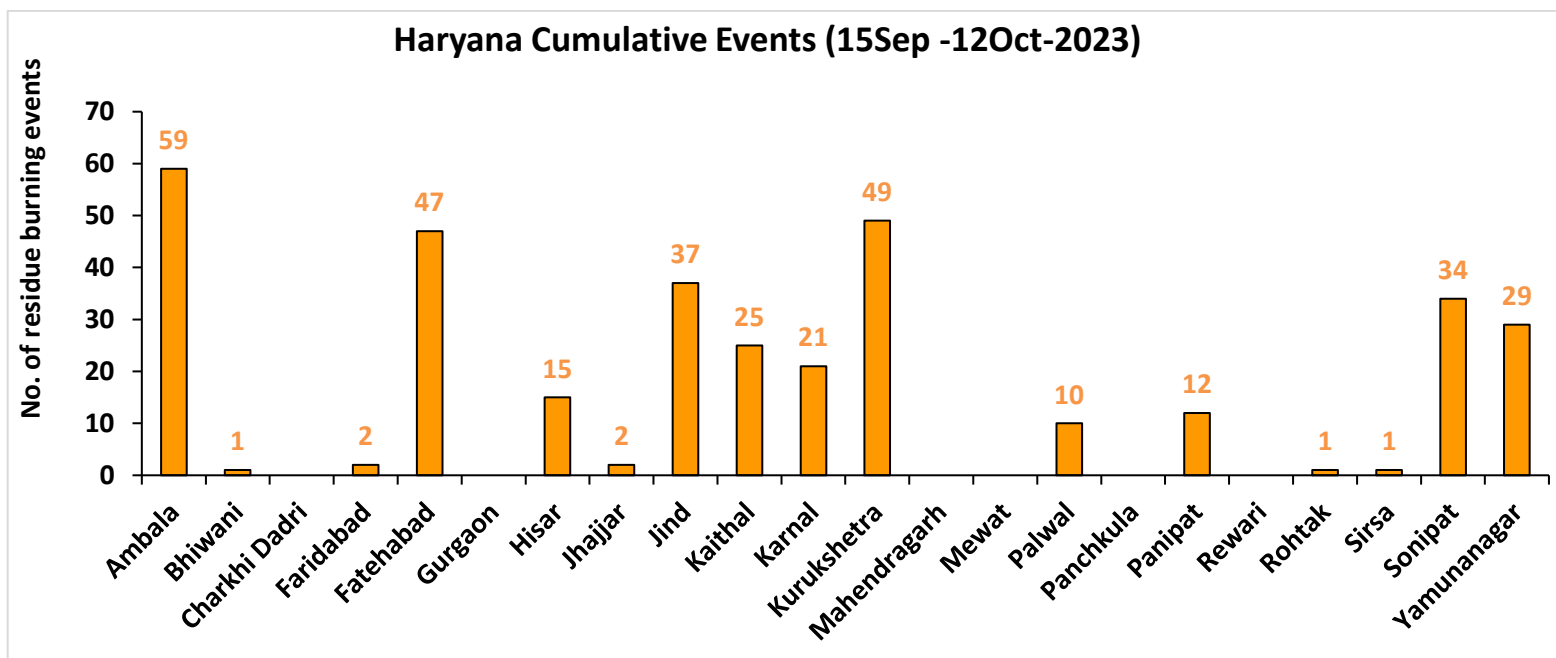
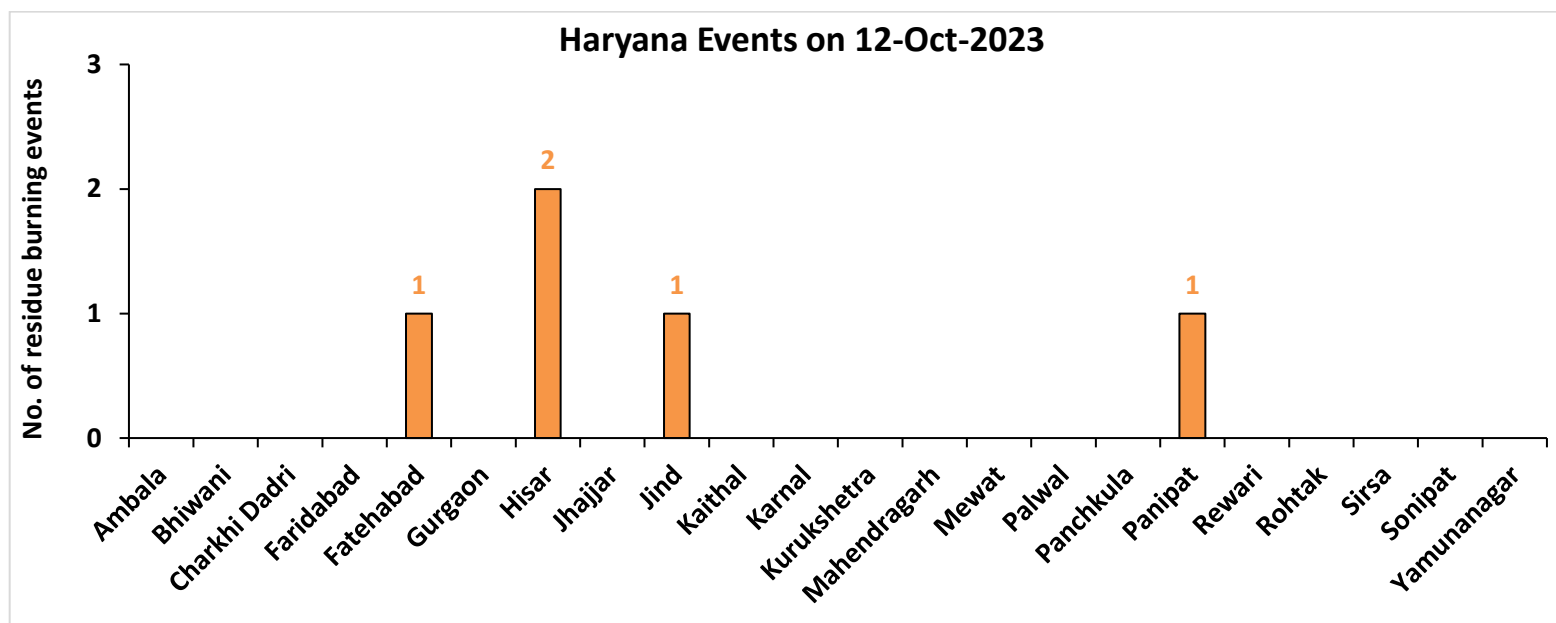
Fire Intensity (W/m²)

- 0 - 5
- 6 - 10
- 11 - 15
- 16 - 20
- >20



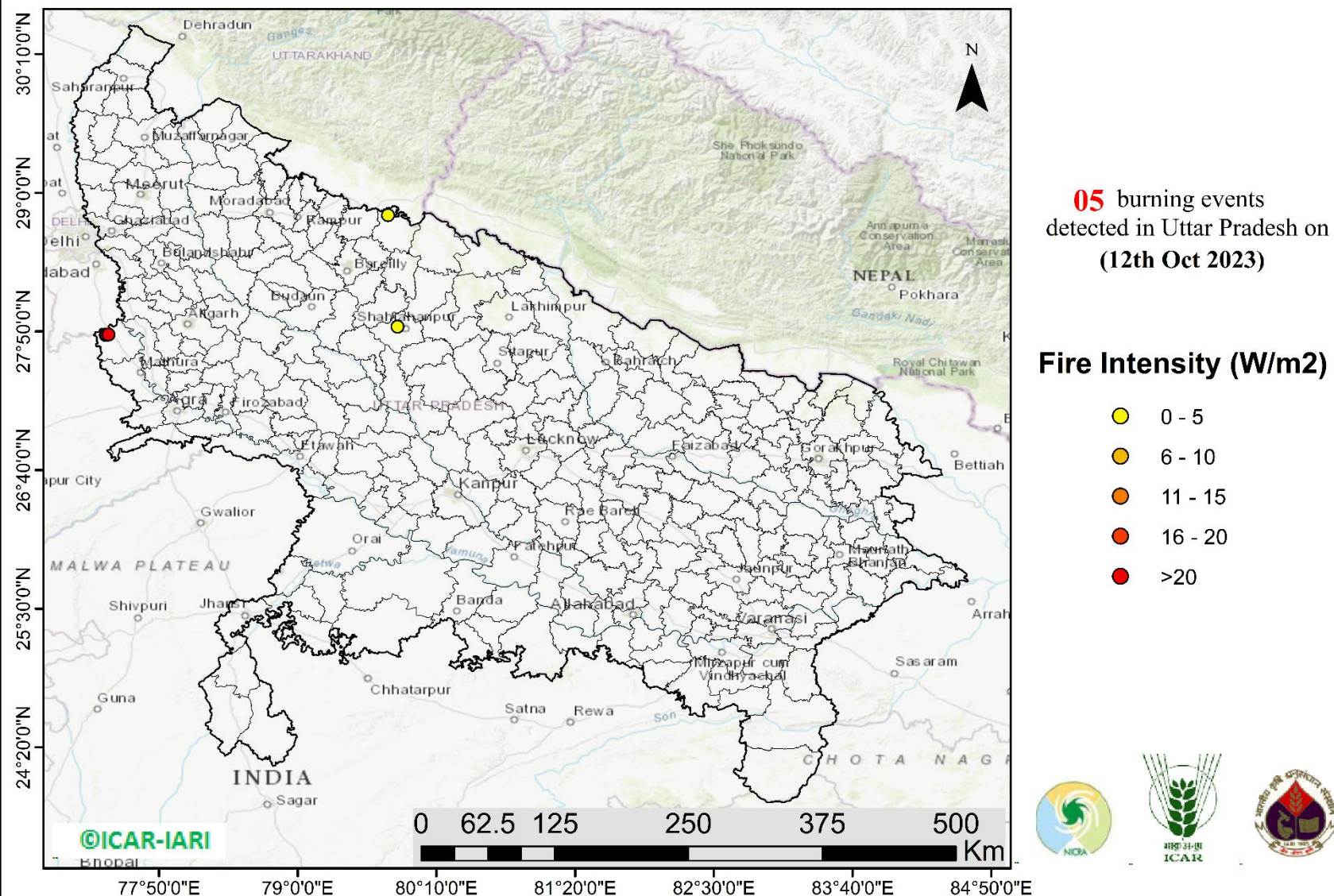
<http://geoportal.icar.gov.in:8080/geoexplorer/composer/>

District-wise cumulative number of residues burning events in Haryana (15Sep-12Oct -2023)



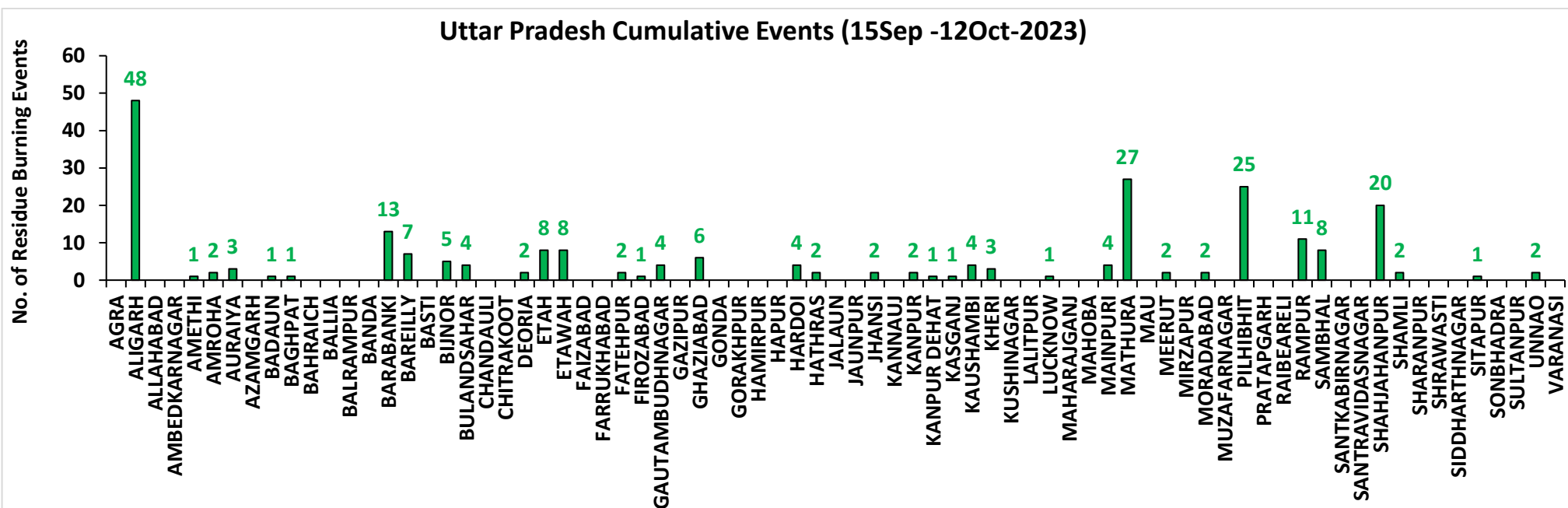
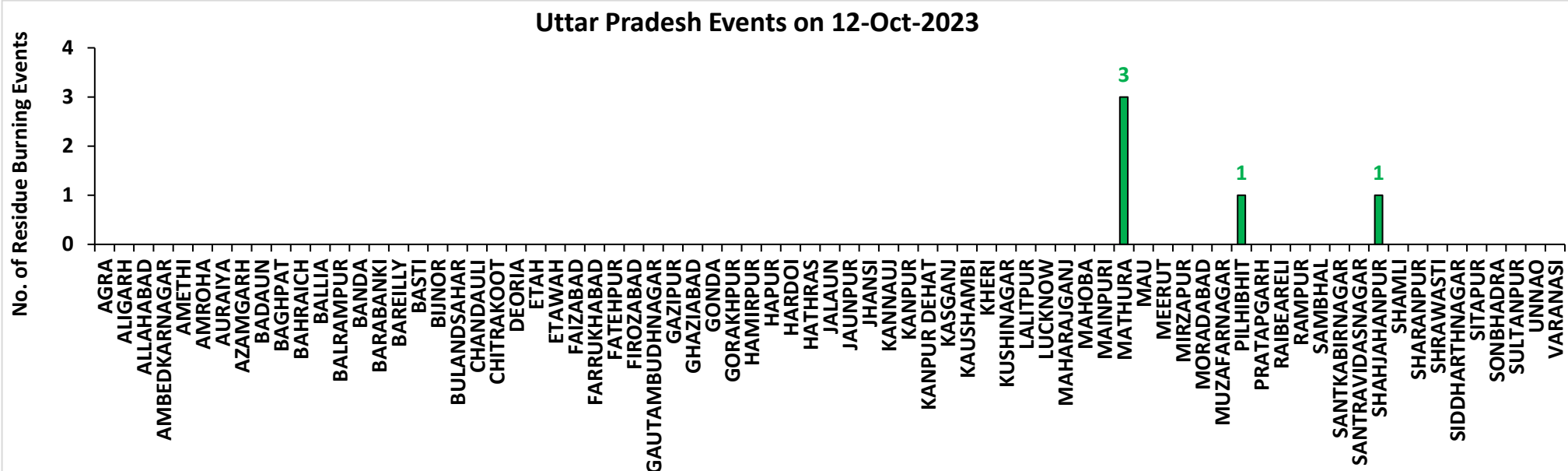
Uttar Pradesh

RICE RESIDUE BURNING IN UTTAR PRADESH



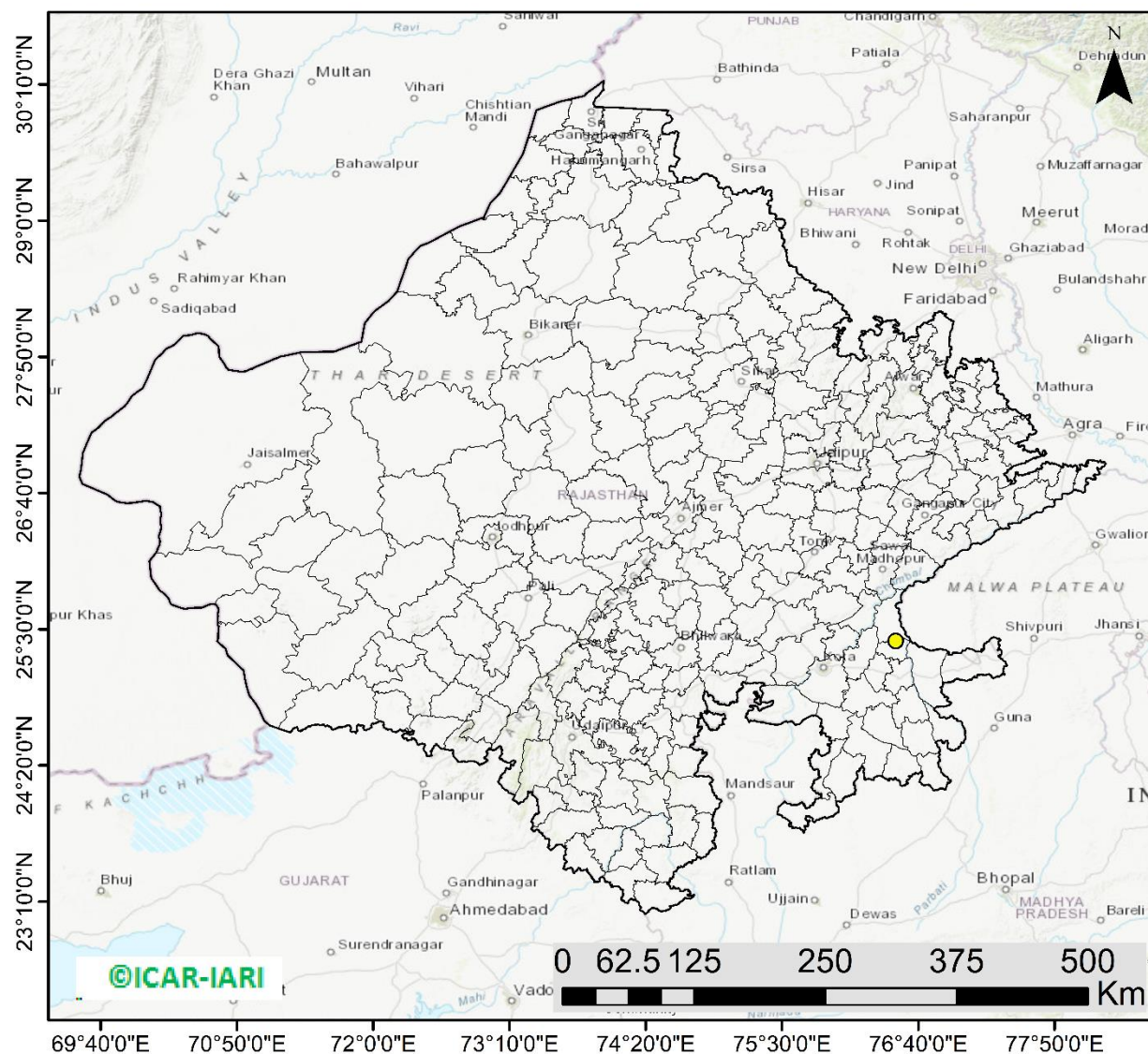
<http://geoportal.icar.gov.in:8080/geoexplorer/composer/>

District-wise cumulative number of residues burning events in Uttar Pradesh (15Sep-12Oct -2023)



Rajasthan

RICE RESIDUE BURNING IN RAJASTHAN



01 burning events
detected in Rajasthan on
(12th Oct 2023)

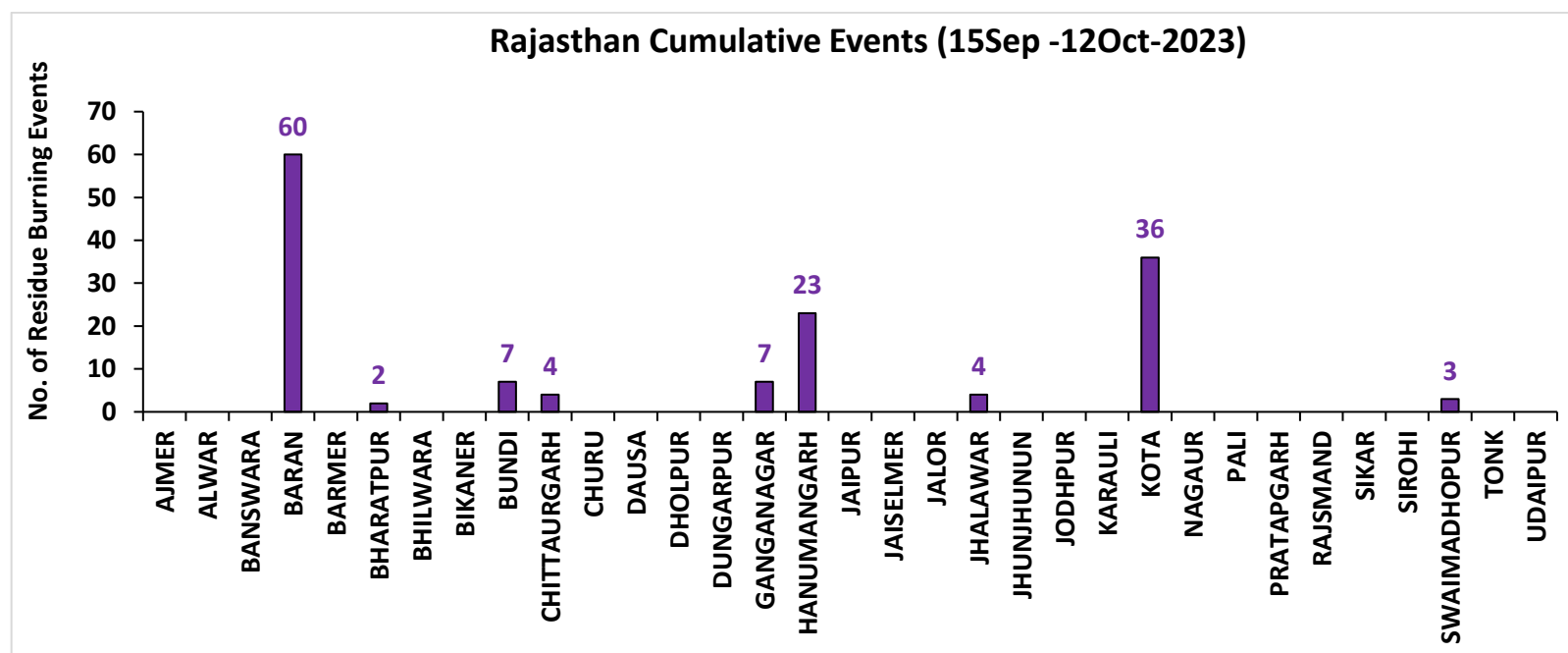
Fire Intensity (W/m²)

- 0 - 5
- 6 - 10
- 11 - 15
- 16 - 20
- >20



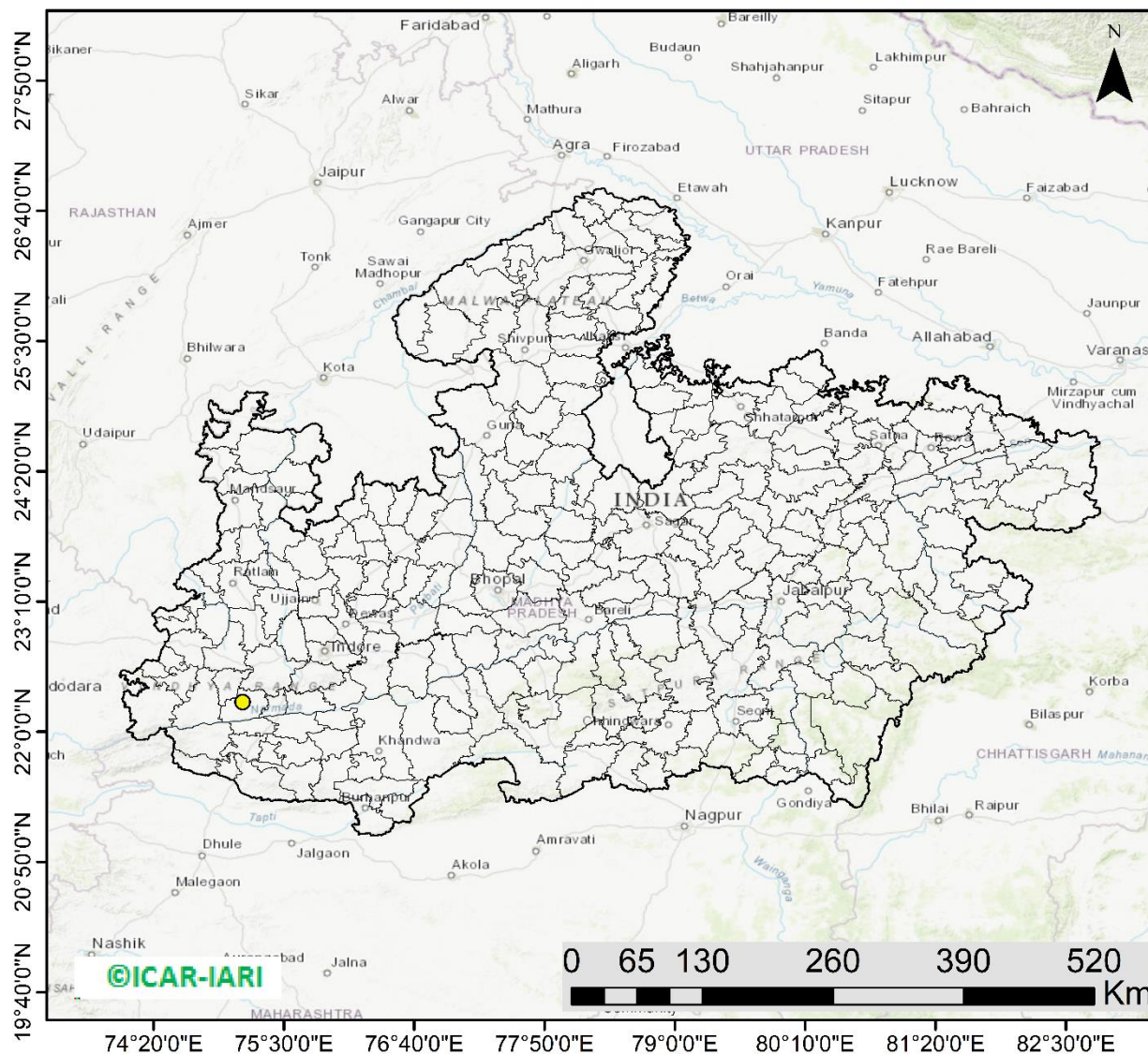
<http://geoportal.icar.gov.in:8080/geoexplorer/composer/>

District-wise cumulative number of residues burning events in Rajasthan (15Sep-12Oct -2023)



Madhya Pradesh

RICE RESIDUE BURNING IN MADHYA PRADESH



01 burning events
detected in MP on
(12th Oct 2023)

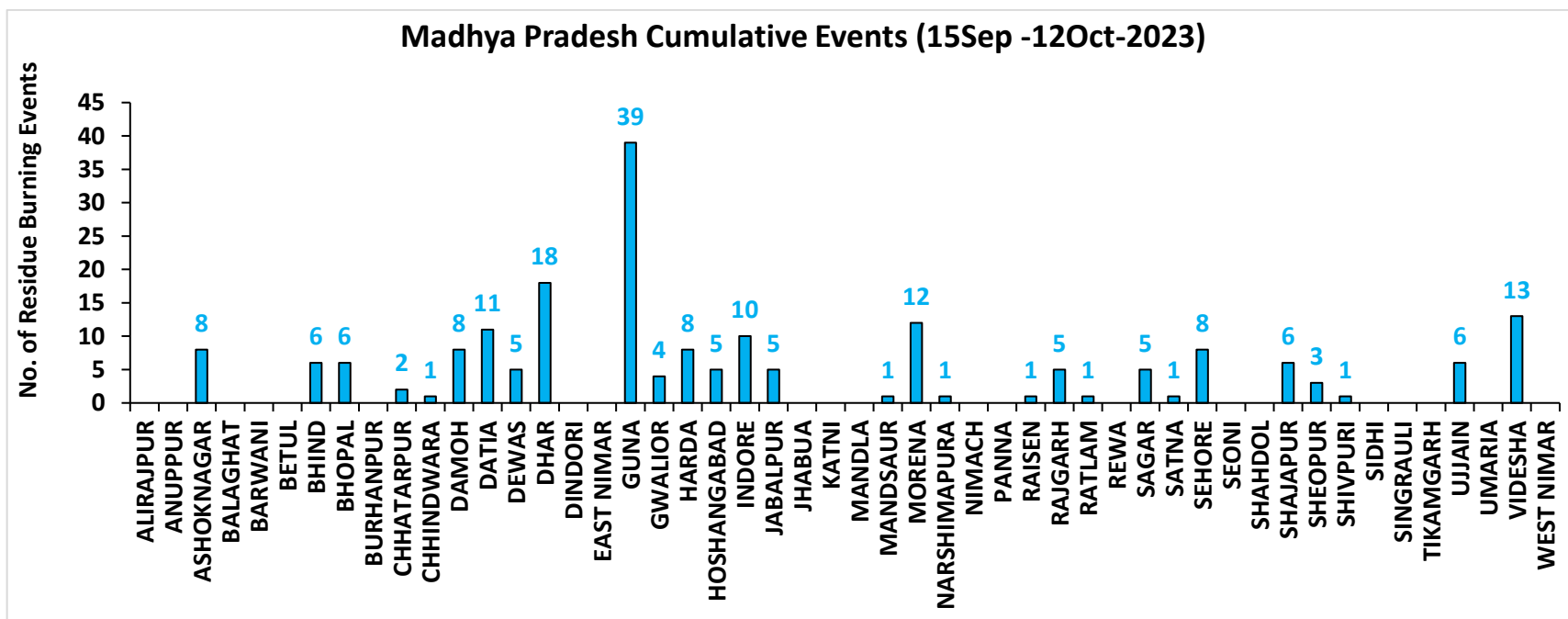
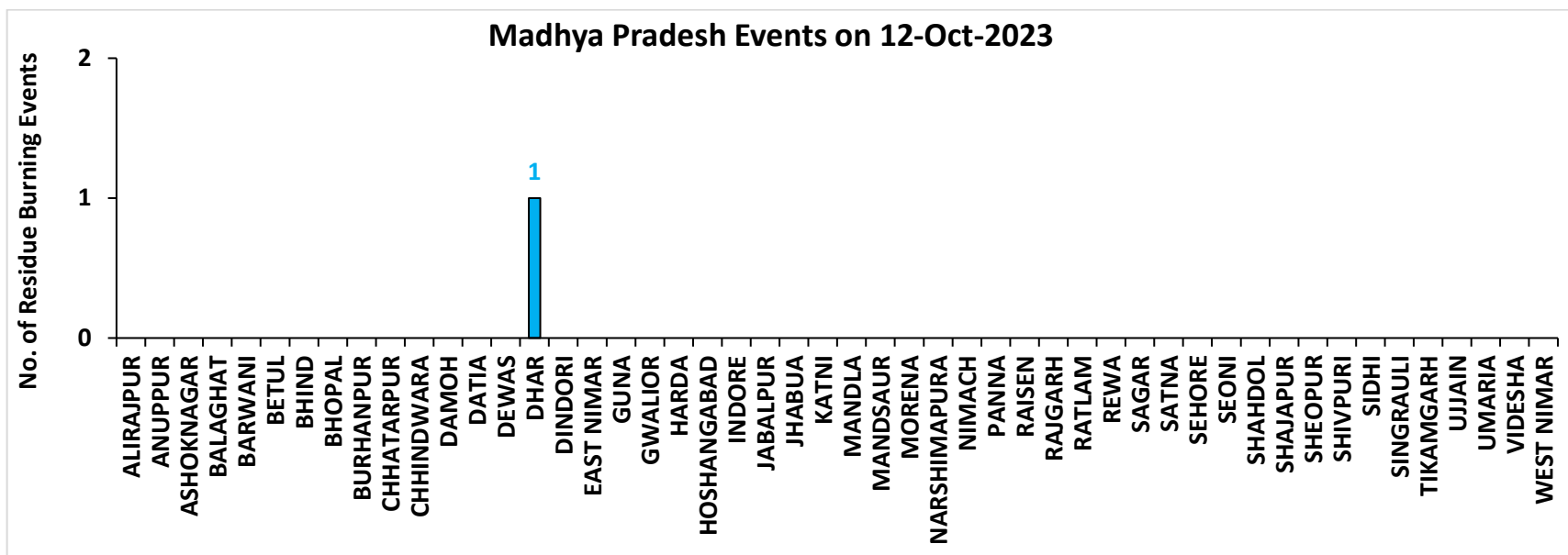
Fire Intensity (W/m²)

- 0 - 5
- 6 - 10
- 11 - 15
- 16 - 20
- >20



<http://geoportal.icar.gov.in:8080/geoexplorer/composer/>

District-wise cumulative number of residues burning events in Madhya Pradesh (15Sep-12Oct -2023)



District-wise cumulative number of residues burning events detected in 2020, 2021, 2022 and 2023
(Period: 15 September-12 October)

(a) Punjab

District	15 Sep-12 Oct			
	2020	2021	2022	2023
Amritsar	1485	355	487	575
Barnala	9	6	6	6
Bathinda	15	0	0	0
Faridkot	44	22	6	8
Fatehgarh Sahib	51	31	4	12
Fazilka	10	7	1	3
Firozpur	132	27	27	32
Gurdaspur	173	33	28	10
Hoshiarpur	8	1	0	2
Jalandhar	46	24	20	16
Kapurthala	74	27	34	62
Ludhiana	93	69	12	10
Mansa	16	1	1	8
Moga	25	8	12	2
Muktsar	6	2	1	0
Pathankot	0	0	0	0
Patiala	256	81	26	93
Rupnagar	10	4	0	1
Sangrur	85	13	12	43
SAS NAGAR (Mohali)	19	13	4	30
SBS Nagar	9	2	2	2
Tarn Taran	779	199	184	161
Total	3345	925	867	1076

(b) Haryana

District	15 Sep-12 Oct			
	2020	2021	2022	2023
AMBALA	65	14	10	59
BHIWANI	0	0	0	1
CHARKHI DADRI	0	0	0	0
FARIDABAD	0	1	0	2
FATEHABAD	31	11	3	47
GURGAON	0	0	0	0
HISAR	6	0	1	15
JHAJJAR	1	0	0	2
JIND	14	9	8	37
KAITHAL	80	61	9	25
KARNAL	131	113	20	21
KURUKSHETRA	94	63	29	49
MAHENDRAGARH	0	0	0	0
MEWAT	0	0	0	0
PALWAL	5	7	0	10
PANCHKULA	0	0	0	0
PANIPAT	7	3	1	12
REWARI	0	0	0	0
ROHTAK	0	0	0	1
SIRSA	5	0	1	1
SONIPAT	8	1	2	34
YAMUNANAGAR	33	15	3	29
Total	480	298	87	345

(c) Uttar Pradesh

District	15 Sep-12 Oct													
	2020	2021	2022	2023	GAUTAMBUDHNAGAR	1	5	0	4	MUZAFARNAGAR	2	0	0	0
AGRA	1	0	0	0	GAZIPUR	0	0	0	0	PILHIBHIT	4	8	2	25
ALIGARH	20	4	1	48	GHAZIABAD	1	2	1	6	PRATAPGARH	0	0	0	0
ALLAHABAD	0	0	0	0	GONDA	0	0	0	0	RAIBEARELI	1	0	0	0
AMBEDKARNAGAR	0	0	0	0	GORAKHPUR	0	0	0	0	RAMPUR	13	16	0	11
AMETHI	0	1	0	1	HAMIRPUR	0	0	0	0	SAMBHAL	6	2	4	8
AMROHA	1	0	0	2	HAPUR	3	0	0	0	SANTKABIRNAGAR	0	0	0	0
AURAIYA	14	1	4	3	HARDOI	2	5	0	4	SANTRAVIDASNAGAR	0	0	0	0
AZAMGARH	0	0	0	0	HATHRAS	0	0	0	2	SHAHJAHANPUR	20	94	53	20
BADAUN	0	0	0	1	JALAUN	0	0	0	0	SHAMLI	18	8	2	2
BAGHPAT	1	0	0	1	JAUNPUR	0	0	0	0	SHARANPUR	18	1	0	0
BAHRAICH	2	2	0	0	JHANSI	6	0	0	2	SHRAWASTI	0	0	0	0
BALLIA	0	0	0	0	KANNAUJ	2	0	0	0	SIDDHARTHANAGAR	0	0	0	0
BALRAMPUR	0	0	0	0	KANPUR	0	1	0	2	SITAPUR	8	1	0	1
BANDA	0	0	0	0	KANPUR DEHAT	2	0	0	1	SONBHADRA	0	0	0	0
BARABANKI	10	0	0	13	KASGANJ	1	0	0	1	SULTANPUR	0	0	0	0
BAREILLY	8	4	0	7	KAUSHAMBI	7	1	0	4	UNNAO	2	0	0	2
BASTI	2	0	0	0	KHERI	23	20	8	3	VARANASI	0	0	0	0
BIJNOR	3	1	1	5	KUSHINAGAR	0	0	0	0	Total	260	201	80	240
BULANDSAHAR	6	1	2	4	LALITPUR	0	1	2	0					
CHANDAULI	0	0	0	0	LUCKNOW	2	0	0	1					
CHITRAKOOT	0	0	0	0	MAHARAJGANJ	0	0	0	0					
DEORIA	0	0	0	2	MAHOBAB	1	0	0	0					
ETAH	4	4	0	8	MAINPURI	3	2	0	4					
ETAWAH	2	3	0	8	MATHURA	6	7	0	27					
FAIZABAD	0	0	0	0	MAU	0	0	0	0					
FARRUKHABAD	0	0	0	0	MEERUT	9	0	0	2					
FATEHPUR	14	0	0	2	MIRZAPUR	5	5	0	0					
FIROZABAD	4	0	0	1	MORADABAD	2	1	0	2					

(d) Delhi

District	15 Sep-12 Oct			
	2020	2021	2022	2023
CENTRAL	0	0	0	0
EAST	0	0	0	0
NEW DELHI	0	0	0	0
NORTH	2	0	2	2
NORTH EAST	0	0	0	0
NORTH WEST DELHI	0	0	0	0
SHAHADRA	0	0	0	0
SOUTH	1	0	0	0
SOUTH EAST	0	0	0	0
SOUTH WEST DELHI	0	0	0	0
WEST DELHI	0	0	0	0
Total	3	0	2	2

(e) Rajasthan

District	15 Sep-12 Oct			
	2020	2021	2022	2023
AJMER	0	0	0	0
ALWAR	1	0	0	0
BANSWARA	0	0	0	0
BARAN	76	0	3	60
BARMER	2	0	0	0
BHARATPUR	0	0	0	2
BHILWARA	2	0	0	0
BIKANER	0	0	0	0
BUNDI	3	0	3	7
CHITTAURGARH	0	1	1	4
CHURU	0	0	0	0
DAUSA	0	0	0	0
DHOLPUR	0	0	0	0
DUNGARPUR	0	0	0	0
GANGANAGAR	2	0	1	7
HANUMANGARH	53	10	3	23
JAIPUR	0	0	0	0
JAISELMER	2	0	0	0
JALOR	0	0	0	0
JHALAWAR	0	0	0	4
JHUNJHUNUN	1	0	0	0
JODHPUR	1	0	0	0
KARAULI	1	0	0	0
KOTA	21	0	1	36
NAGAU	0	0	0	0
PALI	1	0	0	0
PRATAPGARH	0	0	0	0
RAJSMAND	0	0	0	0
SIKAR	0	0	0	0
SIROHI	1	0	0	0
SWAIMADHOPUR	2	0	0	3
TONK	1	0	0	0
UDAIPUR	0	0	0	0
Total	170	11	12	146

(f) Madhya Pradesh

District	15 Sep-12 Oct								
	2020	2021	2022	2023	PANNA	2	0	0	0
ALIRAJPUR	0	0	0	0	RAISEN	10	0	0	1
ANUPPUR	0	0	0	0	RAJGARH	6	1	2	5
ASHOKNAGAR	9	0	0	8	RATLAM	0	0	0	1
BALAGHAT	0	0	0	0	REWA	2	0	0	0
BARWANI	0	0	0	0	SAGAR	13	3	1	5
BETUL	0	0	0	0	SATNA	3	0	0	1
BHIND	0	0	0	6	SEHORE	5	0	1	8
BHOPAL	4	2	0	6	SEONI	6	2	0	0
BURHANPUR	0	0	0	0	SHAHDOL	0	0	0	0
CHHATARPUR	4	0	0	2	SHAJAPUR	5	0	0	6
CHHINDWARA	0	0	0	1	SHEOPUR	5	0	0	3
DAMOH	12	1	0	8	SHIVPURI	4	0	0	1
DATIA	19	2	0	11	SIDHI	0	0	0	0
DEWAS	3	0	0	5	SINGRAULI	0	0	0	0
DHAR	19	0	0	18	TIKAMGARH	1	0	0	0
DINDORI	0	0	0	0	UJJAIN	5	0	0	6
EAST NIMAR	2	1	0	0	UMARIA	0	0	0	0
GUNA	10	1	1	39	VIDESHA	49	5	2	13
GWALIOR	21	0	1	4	WEST NIMAR	2	0	0	0
HARDA	8	7	0	8	Total	254	30	8	200
HOSHANGABAD	9	0	0	5					
INDORE	2	0	0	10					
JABALPUR	6	1	0	5					
JHABUA	0	0	0	0					
KATNI	0	0	0	0					
MANDLA	0	0	0	0					
MANDSAUR	0	0	0	1					
MORENA	7	3	0	12					
NARSHIMAPURA	1	1	0	1					
NIMACH	0	0	0	0					

State-wise and date-wise number of paddy residue burning events detected in 2020, 2021, 2022 and 2023
(Period: 15 September-12 October)

Number of Paddy Residue Burning Events Detected

Punjab					Haryana				Uttar Pradesh			
Date	2020	2021	2022	2023	2020	2021	2022	2023	2020	2021	2022	2023
15-Sep	5	0	14	0	2	0	0	1	1	0	0	0
16-Sep	0	2	0	6	1	0	0	1	0	0	0	2
17-Sep	5	0	1	0	3	0	0	0	0	0	0	0
18-Sep	17	9	7	0	0	0	1	0	0	1	0	0
19-Sep	30	5	18	0	2	0	0	1	0	1	0	0
20-Sep	67	8	10	0	6	0	0	2	2	0	0	0
21-Sep	54	0	56	1	2	0	0	2	1	0	0	1
22-Sep	34	0	30	0	1	0	0	2	1	0	0	0
23-Sep	49	0	3	1	0	0	0	3	1	0	0	0
24-Sep	78	5	0	0	0	0	0	0	1	1	0	1
25-Sep	184	11	0	0	11	0	0	1	5	0	0	1
26-Sep	190	19	0	3	5	0	0	2	8	0	0	0
27-Sep	78	37	0	21	8	0	0	9	4	1	0	0
28-Sep	55	28	0	33	6	1	0	8	1	6	0	4
29-Sep	63	64	0	68	9	4	0	17	6	10	0	6
30-Sep	116	27	8	81	22	2	0	26	13	13	10	18
01-Oct	165	13	45	123	15	4	1	21	7	13	3	17
02-Oct	141	0	83	119	30	2	7	24	12	0	36	7
03-Oct	110	9	75	105	8	6	8	17	4	3	29	8
04-Oct	162	23	65	95	29	2	7	29	8	0	2	9
05-Oct	161	18	130	98	17	3	24	24	14	2	0	14
06-Oct	171	42	85	91	30	0	26	19	12	1	0	18
07-Oct	197	72	62	32	32	11	6	13	41	13	0	5
08-Oct	95	108	19	92	29	27	0	55	25	11	0	27
09-Oct	121	114	3	58	20	39	1	14	15	20	0	13
10-Oct	232	150	4	10	33	95	2	28	9	54	0	34
11-Oct	377	64	45	26	90	51	0	21	23	28	0	50

12-Oct	388	97	104	13	69	51	4	5	46	23	0	5
Total	3345	925	867	1076	480	298	87	345	260	201	80	240

Number of Paddy Residue Burning Events Detected

Delhi					Rajasthan				Madhya Pradesh			
Date	2020	2021	2022	2023	2020	2021	2022	2023	2020	2021	2022	2023
15-Sep	0	0	0	0	1	0	0	0	2	0	0	0
16-Sep	0	0	0	0	0	0	0	0	0	0	0	0
17-Sep	0	0	0	0	0	0	0	0	0	0	0	0
18-Sep	0	0	0	0	0	0	0	0	0	0	0	0
19-Sep	0	0	0	0	0	0	0	0	0	0	0	1
20-Sep	0	0	0	0	0	0	0	0	0	0	0	1
21-Sep	0	0	0	0	1	0	1	0	2	0	0	0
22-Sep	0	0	0	0	0	0	0	0	0	0	0	0
23-Sep	0	0	0	0	1	0	0	0	0	0	0	0
24-Sep	0	0	0	0	0	0	0	0	0	0	0	0
25-Sep	0	0	0	0	1	0	0	0	0	0	0	1
26-Sep	0	0	0	1	1	0	0	0	2	0	0	0
27-Sep	0	0	0	0	0	0	0	0	1	0	0	0
28-Sep	0	0	0	0	0	0	0	0	1	1	0	2
29-Sep	0	0	0	0	3	0	0	6	0	4	0	5
30-Sep	1	0	0	0	1	0	0	4	5	0	0	11
01-Oct	0	0	0	0	4	1	1	4	13	0	1	8
02-Oct	1	0	0	0	7	0	3	4	12	0	1	1
03-Oct	0	0	0	0	9	0	2	5	5	1	5	5
04-Oct	0	0	0	0	15	0	0	22	4	2	0	12
05-Oct	0	0	2	0	14	0	1	11	13	1	0	18
06-Oct	1	0	0	0	16	0	0	9	37	1	0	7
07-Oct	0	0	0	0	14	1	1	7	31	1	0	9
08-Oct	0	0	0	0	11	0	0	9	17	2	0	18
09-Oct	0	0	0	0	12	0	0	17	15	4	0	24
10-Oct	0	0	0	0	14	1	0	26	18	10	0	52
11-Oct	0	0	0	1	21	7	1	21	32	2	0	24

12-Oct	0	0	0	0	24	1	2	1	44	1	1	1
Total	3	0	2	2	170	11	12	146	254	30	8	200

Details of residue burning events in Punjab on 12-Oct-2023

S. No.	District	Tehsil / Block	Satellite	Latitude	Longitude	ACQ_TIME	Day / Night	Fire Power (W/m2)
1	AMRITSAR	JANDIALA GURU	S-NPP	31.57597	75.05508	14:31:47	D	3.30
2	AMRITSAR	JANDIALA GURU	S-NPP	31.59789	74.98389	14:31:47	D	7.70
3	AMRITSAR	JANDIALA GURU	S-NPP	31.60219	74.98309	14:31:47	D	6.20
4	FIROZPUR	FIROZPUR	S-NPP	30.98279	74.58535	14:31:47	D	4.60
5	FIROZPUR	GHALL KHURD	S-NPP	30.90488	74.71472	14:31:47	D	9.80
6	FIROZPUR	GHALL KHURD	S-NPP	30.90516	74.72214	14:31:47	D	9.80
7	FIROZPUR	MAMDOT	S-NPP	30.78883	74.38897	14:31:47	D	14.40
8	KAPURTHALA	SULTANPUR LODHI	S-NPP	31.27057	75.20098	14:31:47	D	7.50
9	LUDHIANA	MACHHIWARA	AQUA	30.87600	76.27700	13:24:23	D	18.40
10	SANGRUR	MALERKOTLA-2	S-NPP	30.57743	75.89625	14:31:47	D	4.30
11	TARN TARAN	KHADUR SAHIB	S-NPP	31.41638	75.08392	14:31:47	D	4.20
12	TARN TARAN	TARN TARAN	S-NPP	31.44539	74.78706	14:31:47	D	8.80
13	TARN TARAN	TARN TARAN	S-NPP	31.45009	74.78079	14:31:47	D	9.50

Details of residue burning events in Haryana on 12-Oct-2023

S. No.	District	Tehsil / Block	Satellite	Latitude	Longitude	ACQ_TIME	Day / Night	Fire Power (W/m2)
1	FATEHABAD	TOHANA	S-NPP	29.70628	75.79124	14:31:47	D	4.90
2	HISAR	BARWALA	S-NPP	29.42802	76.04473	14:31:47	D	6.30
3	HISAR	BARWALA	S-NPP	29.42878	76.04613	14:31:47	D	6.00
4	JIND	NARWANA	S-NPP	29.54594	76.04039	14:31:47	D	8.80
5	PANIPAT	MADLAUDA	S-NPP	29.43390	76.76456	01:41:30	N	0.70

Details of residue burning events in Uttar Pradesh on 12-Oct-2023

S. No.	District	Tehsil / Block	Satellite	Latitude	Longitude	ACQ_TIME	Day / Night	Fire Power (W/m2)
1	MATHURA	CHHATA	AQUA	27.80200	77.38700	13:24:23	D	31.00
2	MATHURA	CHHATA	AQUA	27.80600	77.39200	13:24:23	D	48.70
3	MATHURA	CHHATA	AQUA	27.80600	77.41000	13:24:23	D	31.20
4	PILHIBHIT	PILHIBHIT	S-NPP	28.81022	79.76006	01:41:30	N	0.80
5	SHAHJAHANPUR	SHAHJAHANPUR	S-NPP	27.87223	79.84023	01:41:30	N	0.50

Details of residue burning events in Rajasthan on 12-Oct-2023

S. No.	District	Tehsil / Block	Satellite	Latitude	Longitude	ACQ_TIME	Day / Night	Fire Power (W/m2)
1	BARAN	MANGROL	S-NPP	25.39821	76.47401	14:31:47	D	3.70

Details of residue burning events in Madhya Pradesh on 12-Oct-2023

S. No.	District	Tehsil / Block	Satellite	Latitude	Longitude	ACQ_TIME	Day / Night	Fire Power (W/m2)
1	DHAR	MANAWAR	S-NPP	22.26510	75.13140	01:41:30	N	1.80

The study used data received from following Satellites at NRSC Ground Station

S. No.	Satellite Name	Sensor	Resolution (meter)	Day / Night Passes
1.	Suomi NPP	VIIRS	375 / 750	Both
2.	Terra	MODIS	1000	Both
3.	Aqua	MODIS	1000	Both

**The rice mask for the Punjab and Haryana were provided by MNCFC, MoA&FW, Govt of India
The kharif crop mask for Delhi, Uttar Pradesh, Rajasthan and Madhya Pradesh were provided by NRSC (ISRO)**

Study Team

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GIS Maps of fire events can be visualized online on ICAR KRISHI Geoportal website: <http://geoportal.icar.gov.in:8080/geoexplorer/composer/>
(Part of KRISHI Portal: <https://krishi.icar.gov.in> initiative)
